

PROTECTED TAILS AND POLYNOMIAL-TIME ENUMERATION OF PERMUTATIONS AVOIDING A DIRECT SUM OF AN INCREASING PATTERN AND 231

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ABSTRACT. We give an algorithm counting the permutations that avoid a fixed pattern of the following form: the direct sum of an increasing pattern and 231. The first members of the family are 1342 and 12453. For each member the algorithm uses polynomially many operations and stored integers, with degrees that grow linearly in the length of the pattern. It comes from a recurrence that reads a permutation from left to right and records the constraints that the letters read so far impose on those still unread. This recurrence has exponentially many states, but part of each state is protected: later steps carry it along unchanged and do not depend on it, and factoring the protected part out leaves a dynamic program of polynomial size. For 12453 a translation symmetry sharpens the bounds to degree seven for the operations and degree four for the storage. We also compute the number of 12453-avoiding permutations of every length up to 150. The previously published series, due to Biers-Ariel (2019), reached length 38. We also generate uniformly random avoiders, which we illustrate with a heatmap of one million 12453-avoiding permutations of length 300. The counting recurrences for 1342 and 12453 are verified in the Lean 4 proof assistant.

1. INTRODUCTION

A permutation $\pi = \pi_1 \cdots \pi_n$ of $[n] = \{1, \dots, n\}$ *contains* a pattern τ in the symmetric group S_k if there are indices $1 \leq i_1 < \cdots < i_k \leq n$ for which $\pi_{i_1}, \dots, \pi_{i_k}$ has the same relative order as τ . Otherwise π *avoids* τ . For example, the entries 2, 5, 7, 9, 6 in 251796834 have relative ranks 1, 2, 4, 5, 3, and therefore form an occurrence of 12453. We write $\text{Av}_n(\tau)$ for the set of τ -avoiding permutations of $[n]$, and $\text{Av}(\tau) = \bigcup_{n \geq 0} \text{Av}_n(\tau)$. Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$.

The basic enumerative problem is to determine

$$a_n(\tau) = |\text{Av}_n(\tau)|.$$

By *polynomial-time enumeration* we mean computing this integer, not listing the avoiding permutations. In other words, we seek a *Wilfian formula* [25, 26]. For a fixed pattern, the theorem of Marcus and Tardos [18] gives an exponential upper bound $a_n(\tau) \leq c(\tau)^n$ on the counting sequence. Arratia [4] showed that $a_n(\tau)^{1/n}$ converges. Its limit, which is finite by this bound, is called the *Stanley–Wilf limit* of τ , or the *growth rate* of $\text{Av}(\tau)$. The task is to obtain $a_n(\tau)$ in at most $P(n)$ steps, where P is a polynomial. No general principle can apply, since Garrabrant and Pak [15] constructed a finitely based class whose counting sequence cannot be computed in polynomial time unless $\text{EXP} = \oplus \text{EXP}$.¹

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¹Here EXP is the class of decision problems solvable in deterministic exponential time, $\oplus \text{EXP}$ is the class of problems that ask whether a nondeterministic exponential-time machine has an odd number of accepting computations, and the two classes are widely believed to differ.

If $\alpha \in S_r$ and $\beta \in S_s$, their *direct sum* $\alpha \oplus \beta$ is the word obtained by writing α and then writing β with every value increased by r . We will focus on the following direct-sum family. Put

$$\iota_d = 12 \cdots d, \quad \beta_d = \iota_d \oplus 231 = 12 \cdots d(d+2)(d+3)(d+1). \quad (1)$$

Thus

$$\beta_1 = 1342, \quad \beta_2 = 12453, \quad \beta_3 = 123564, \quad \beta_4 = 1234675.$$

The case $d = 1$ is already enumerated algebraically by Bóna [9]. The family itself also appears in Bóna [10]: from $\beta_d = 1 \oplus \beta_{d-1}$ for $d \geq 2$, and $\beta_1 = 1 \oplus 231$, he obtained a recursive deletion rule for *left-to-right minima* (entries smaller than every entry to their left) and used it to determine the Stanley–Wilf limits. That rule does not by itself determine the exact counts from the unrefined β_{d-1} counts. The next case is $d = 2$, giving $\beta_2 = 12453$. The counting sequence begins

$$1, 1, 2, 6, 24, 119, 694, 4581, 33286, 260927, 2174398, 19053058, \dots \quad (2)$$

In the census of Clisby, Conway, Guttmann, and Inoue [12], the 12453 sequence was known through $n = 38$ from a specialized program of Biers-Ariel [7, 8]. The program uses a composition-valued state of unbounded length, and the number of its states grows exponentially. Our starting point is the same recurrence: for $d = 2$ the literal recurrence developed below coincides with Biers-Ariel’s, state for state, and we extend it to the whole family β_d . We then show how to shrink its exponential state space to polynomial size.

The family also has a description by a sorting device. Albert, Homberger, Pantone, Shar, and Vatter [2] study the $\mathcal{C}$ -machine, which pushes each input entry anywhere into a container whose contents must remain in the class $\mathcal{C}$ and pops the container from its left end. Their Theorem 1.1 shows that the $\text{Av}(B)$ -machine generates exactly $\text{Av}(1 \ominus B)$, and they observe that $\text{Av}(4213)$, the complement of $\text{Av}(1342)$, is generated by the $\text{Av}(213)$ -machine, answering a question of Bóna. Since $\beta_d^c = 1 \ominus \beta_{d-1}^c$ for the complement β_d^c , the class $\text{Av}(\beta_d)$ is, up to complementation, the class generated by the $\text{Av}(\beta_{d-1}^c)$ -machine. For 12453 the container class is $\text{Av}(4213)$. They prove that the generated class has a rational generating function when $\mathcal{C}$ is finite, an algebraic one when $\mathcal{C}$ is bounded, that is, when $|\mathcal{C}_n|$ is bounded by a constant, and a polynomial-time counting algorithm when $|\mathcal{C}_n|$ is bounded by a polynomial in n . The container class $\text{Av}(4213)$ has growth rate 8, so none of these conditions applies to 12453.

Suppose π avoids β_d . Every occurrence of $12 \cdots d$ in a prefix of π forces the unread values above it to be read in a 231-avoiding order, and we record the conditions created by a prefix as an ordered stack of intervals of unread values (section 2). Reading one more letter changes the control, that is, the counts of unread values that no condition restricts, or the active head of the stack, or both. The intervals below the head are not touched until the head has been consumed, and the transitions up to that point do not depend on them. A transfer kernel records the control at which this happens. This removes the stack from the memoization key, and the resulting table has polynomial size.

Our main result is the following. The notation $O_d(\cdot)$ allows the hidden constant to depend on the fixed family parameter d .

Theorem 1.1. *For every fixed $d \geq 1$ and every $N \geq 0$, the numbers*

$$|\text{Av}_0(\beta_d)|, \dots, |\text{Av}_N(\beta_d)|$$

can be computed using $O_d(N^{3d+2})$ integer arithmetic operations and $O_d(N^{2d+1})$ stored integers. All integers involved have $O_d(N \log N)$ bits. Therefore the bit complexity² is polynomial. More precisely it is

$$O_d\left(N^{3d+2}M(N \log N)\right),$$

²The operation count treats each integer addition or multiplication as one step. The bit complexity counts operations on single bits, so the sizes of the integers enter through the multiplication cost $M(b)$ defined below.

where $M(b)$ is the cost of multiplying b -bit integers, and the bit storage is $O_d(N^{2d+2} \log N)$.

In particular, the family theorem gives time $O(N^5)$ and storage $O(N^3)$ for $\text{Av}(1342)$, and $O(N^8)$ and $O(N^5)$ for $\text{Av}(12453)$. An additional translation symmetry, proved in [section 7](#), improves this to $O(N^7)$ and $O(N^4)$ for $\text{Av}(12453)$ ([corollary 7.4](#)). The theorem gives a polynomial bound for each fixed d , not a uniform one when d is part of the input.

To our knowledge, no polynomial-time enumeration algorithm was previously available for the family $\text{Av}(\iota_d \oplus 231)$ at every fixed d , or even for the single class $\text{Av}(12453)$. In this paper we show that the literal recurrence is a pushdown system: every transition acts on the interval stack through its active head alone, so the completion counts factor through transfer kernels ([theorem 2.17](#) and [corollary 5.1](#)), although the number of reachable stack states grows like the Fibonacci numbers ([proposition 4.5](#)). The factorization itself is the counting form of the classical conversion of a pushdown automaton into a grammar ([remark 2.18](#)). What is new is that it applies with a control set that grows with n , where it gives a transfer table of polynomial size rather than a finite grammar. We prove the family bounds of [theorem 1.1](#) and two translation quotients for 12453 ([lemmas 7.2](#) and [7.3](#)). We also provide the values $|\text{Av}_n(12453)|$ for $n \leq 150$, computed by Chinese remaindering with a proved bound and checked by separately written implementations ([table 1](#)). The transfer tables can also be used to generate uniformly random avoiders in polynomial time ([proposition 8.1](#)), and the counting recurrences for $d \leq 2$ are machine-checked in Lean 4 ([section 9](#)).

The paper is organized as follows. In [section 2](#) we describe the conditions that a scanned prefix imposes on its completions, represent them by an interval stack, and develop the whole method for $d = 1$, where $\text{Av}(1342)$ serves as a prototype: the literal recurrence, the protected-tail factorization of [theorem 2.17](#), and the resulting polynomial algorithm. In [section 3](#) we replace the single minimum by the d patience-sorting thresholds and their bands. In [section 4](#) we derive the literal recurrence for every d and show that its state space is exponential. In [section 5](#) we apply the factorization to it and obtain the transfer kernels. In [section 6](#) we bound their supports and prove [theorem 1.1](#). In [section 7](#) we specialize to 12453 and use two translation symmetries to reach $O(N^7)$ operations and $O(N^4)$ storage. In [section 8](#) we compute the values $|\text{Av}_n(12453)|$ for $n \leq 150$ and describe the uniform sampler. In [section 9](#) we describe the Lean development. In [section 10](#) we place the class among the length-five Wilf classes and state a conjecture on its asymptotics.

2. TRIGGERS, STACKS, AND THE COMPLETE ONE-THRESHOLD PROTOTYPE

We first describe the restrictions that a scanned prefix imposes on its completions and show that they admit an interval-stack representation for every $d \geq 1$. We then develop the counting algorithm completely for $d = 1$, which isolates the protected-tail construction before the additional threshold coordinates are introduced. The lemmas on triggers and stacks ([lemmas 2.1, 2.2, 2.5](#) and [2.8](#) and [proposition 2.10](#)) are used unchanged for the general family in [section 4](#), and [theorem 2.17](#) is stated for recurrences on a control set and a stack, so that it applies verbatim in [section 5](#).

For a word w with distinct entries, its *standardization* $\text{st}(w)$ is obtained by replacing the smallest entry by 1, the next smallest by 2, and so on. If X is a set of values, $w|_X$ denotes the subword formed by retaining only the entries belonging to X . An *increasing d -subsequence* of a permutation π , or more generally of a word π with distinct entries, is a choice of positions $i_1 < \dots < i_d$ with $\pi_{i_1} < \dots < \pi_{i_d}$. We call its last entry a *d -trigger*. The positions need not be adjacent.

Lemma 2.1 (Trigger lemma). *A permutation $\pi = \pi_1 \cdots \pi_n$ avoids β_d if and only if, for every increasing d -subsequence ending at $c = \pi_j$, the word*

$$\pi_{j+1} \cdots \pi_n|_{\{x: x > \pi_j\}}$$

avoids 231 after standardization.

Proof. Suppose π contains β_d . The first d entries of an occurrence form an increasing d -subsequence, and its last three entries form a 231 whose values are all larger. Conversely, an increasing d -subsequence followed by such a 231 has relative order $\iota_d \oplus 231 = \beta_d$. $\square$

When a trigger c is read, its entire subsequent projection onto values larger than c must avoid 231. Later in the scan, some letters of that projection have already been read. It is not enough to test the unread part of the projection, since an occurrence of 231 may use both read and unread letters. The interval stack below records the resulting conditions on the unread letters.

More formally, the 231-obligation created by a trigger $c = \pi_j$ requires that $\pi_{j+1} \cdots \pi_n|_{\{x: x > c\}}$ avoid 231 after standardization. Let $\sigma = \sigma_1 \cdots \sigma_k$ be the scanned prefix. The letters of $[n]$ not in σ are *unread*, and a *completion* of σ is a word w using each unread letter once. For a trigger $c = \sigma_j$ of σ , its *residual obligation* is the restriction of this requirement to completions of σ , with the already scanned letters of the projection fixed: w satisfies it when $(\sigma_{j+1} \cdots \sigma_k w)|_{\{x: x > c\}}$ avoids 231 after standardization.

2.1. The Catalan interval stack. The following first-letter decomposition of $\text{Av}(231)$ applies to an arbitrary finite ordered alphabet.

Lemma 2.2 (First-letter lemma). *Let I be a finite totally ordered set of size ℓ , and let $x \in I$ have local rank $r = |\{y \in I : y \leq x\}|$. A permutation of I beginning with x avoids 231 if and only if all values below x occur before all values above x , and the two induced subwords avoid 231. Thus reading x replaces an interval of size ℓ by intervals of sizes $r - 1$ and $\ell - r$, in that order, with zero parts deleted.*

Proof. Suppose an upper value occurs before a lower value after x . Then x and these two values form a 231. Conversely, if the lower and upper subwords occur in that order, a 231 cannot use letters from both: both position and value increase from the first block to the second. Nor can a 231 use x and two letters from just one block: x is larger than every lower letter and smaller than every upper letter, while the first entry of a 231 has middle rank. If it uses x , one lower letter, and one upper letter, their order is 213. Avoidance therefore reduces to avoidance inside the two blocks. $\square$

We write $\text{Av}(231)(I)$ for the 231-avoiding words using each value of I once. If $I < J$, that is, every element of I is smaller than every element of J , and u and w are words on I and J , then $u \oplus w$ is the concatenation uw . For sets of such words, $\oplus$ denotes the set of all concatenations. Consider the projection created by one trigger c , and read its letters one at a time. By lemma 2.2, the first letter x splits the rest of the projection into a lower and an upper interval, and every letter of the lower interval must be read before any letter of the upper one. The next letter of the projection therefore lies in the lower interval while that interval is nonempty, and the lemma, applied to the subword on that interval, splits it in turn. The letter itself then plays no further role, since both conditions of the lemma concern only the letters after it. Repeated application of lemma 2.2 thus represents the residual obligation of c by an ordered interval stack

$$I_1 < I_2 < \cdots < I_s,$$

which we also write $I_1 | I_2 | \cdots | I_s$. That is, the unread letters of the projection must form a word in

$$\text{Av}(231)(I_1) \oplus \text{Av}(231)(I_2) \oplus \cdots \oplus \text{Av}(231)(I_s). \quad (3)$$

The read letters of the projection do not appear in (3), and none of them is needed again. Note that an occurrence of 231 involving a read letter x and an unread letter must begin with x : if a letter $a < x$ read earlier were its first entry, then every value of the projection below a would already have been read, because the lower interval of a precedes its upper one, which contains x . Beginning with x , the occurrence needs an upper letter followed by a lower one, which the split at x excludes. For example, relabel the values of a projection as $1, \dots, 6$ and suppose that its first two letters are 4 and 2. Reading 4 leaves the stack $\{1, 2, 3\} \mid \{5, 6\}$, and reading 2 then leaves $\{1\} \mid \{3\} \mid \{5, 6\}$. The completion 1365 is allowed, while a completion beginning 31 is excluded: together with the read letter 2 it would form a 231.

The stack records the obligations created by the scanned prefix. Further obligations arise as the completion is read, and they are incorporated by the stack updates described below. Here and throughout, an *interval* is a consecutive block of the ordered set of currently unread values. Its elements need not be consecutive integers. We call I_1 the *active head* and $(I_2, \dots, I_s)$ the *deferred tail*. Only the active head may be read next among the values of the stack: every letter of I_1 must be read before any letter of I_2 , and so forth. Unread values below the trigger are constrained by no obligation created so far and may be interspersed. For an interval of size ℓ , an *endpoint choice* has local rank 1 or ℓ , and an *interior choice* has local rank in $\{2, \dots, \ell - 1\}$.

We call the gap in value between two consecutive intervals of the current stack a *separation*, and a read letter lying in that gap a *separator*. A separation may have several separators. An interior choice creates a new separation, at the letter read, while an endpoint choice leaves at least one side empty and creates none, although the letter read may lie in an existing separation. Inductively, as long as no trigger smaller than c has been read (a trigger larger than c is itself a letter of the projection and only splits it further), two unread letters $u < v$ of the projection lie in different intervals if and only if some letter y with $u < y < v$ has been read from the projection. [Lemma 2.11](#) and [corollary 4.4](#) below determine the separations of the stack once the obligations of several triggers have been combined.

Example 2.3 (Reading one projection). The permutation

$$\pi = 9, 11, 10, 14, 5, 12, 6, 2, 3, 8, 4, 7, 1, 13, 15 \quad (4)$$

of [15] avoids 1342, and therefore also 12453, and serves as the running example of this paper. For $d = 1$ every letter is a trigger. The projection created by the first letter 9 is the word 11, 10, 14, 12, 13, 15 formed by the later letters above 9. It standardizes to 215346 and avoids 231, as [lemma 2.1](#) requires. We read this projection letter by letter. The stack starts as the single interval $\{10, \dots, 15\}$. The first letter 11 has local rank 2, an interior choice, so [lemma 2.2](#) requires the lower interval $\{10\}$ to be read before the upper interval $\{12, \dots, 15\}$. The stack is $\{10\} \mid \{12, \dots, 15\}$. Reading 10 is an endpoint choice that exhausts the head and leaves $\{12, \dots, 15\}$, and the interior choice 14, of local rank 3 in that interval, splits it into $\{12, 13\} \mid \{15\}$. The unread letters of the projection must now form a word in $\text{Av}(231)(\{12, 13\}) \oplus \text{Av}(231)(\{15\})$: either order of 12 and 13, then 15. The read letters 11, 10, 14 are not retained. While 10 was unread, 11 was the only separator of the separation between $\{10\}$ and $\{12, \dots, 15\}$. Now 14 is the only separator of the separation between $\{12, 13\}$ and $\{15\}$, and 10 and 11 separate nothing.

Several triggers impose overlapping obligations. We now define the object that records all of them: a read prefix together with a stack, updated by three kinds of move. [Lemma 2.8](#) and [proposition 2.10](#) then show that the stack of a legal prefix represents exactly the conjunction of the residual obligations of its triggers, and that the legal prefixes are exactly the prefixes of β_d -avoiding permutations.

Definition 2.4 (Scan states and legal moves). Fix $d \geq 1$ and n . A *scan state* is a pair (σ, S) . Here σ is a word with distinct letters from $[n]$, called the *read prefix*, and the letters of $[n]$ not in σ

are *unread*. We write q for the least d -trigger of σ when σ has one. The *stack* $S = (I_1, \dots, I_s)$ is a list of nonempty sets with $I_1 < I_2 < \dots < I_s$ whose union is the set of unread letters larger than q , so that each I_j is an interval in the sense of [section 2.1](#), I_1 is the active head, and $(I_2, \dots, I_s)$ is the deferred tail. When σ has no d -trigger, S is the empty list, $s = 0$. The *initial state* is $(\varepsilon, ())$: the empty prefix and the empty stack. Let (σ, S) be a scan state and let x be an unread letter. Reading x is a *legal move*, leading to the state $(\sigma x, S')$, in the following three cases.

- (a) *Merger*: x is a d -trigger of σx and is smaller than every d -trigger of σ . Let E be the set of unread letters strictly between x and q , or the set of all unread letters larger than x when σ has no d -trigger. Then $S' = (E \cup I_1, I_2, \dots, I_s)$ if $s \geq 1$, and $S' = (E)$ if $s = 0$, the empty stack when $E = \emptyset$.
- (b) *Split*: x is a d -trigger of σx that exceeds some d -trigger of σ , and x lies in the active head I_1 , so that $s \geq 1$. Then $S' = (I_1^{<x}, I_1^{>x}, I_2, \dots, I_s)$, where $I_1^{<x}$ and $I_1^{>x}$ are the sets of elements of I_1 below and above x , with empty sets deleted. The move is an endpoint or an interior choice, as defined above, depending on the local rank of x in I_1 .
- (c) *Non-trigger move*: x is not a d -trigger of σx . Then $S' = S$. For $d = 1$ every letter is a 1-trigger, so this case does not occur.

Reading any other unread letter is *illegal*. A word σ is *legal* if the initial state reaches a state (σ, S) by legal moves, and we write $\mathcal{S}(\sigma)$ for the resulting stack, the *stack of* σ . Finally, $\mathcal{R}(\sigma)$ denotes the set of permutations of $[n]$ with prefix σ all of whose prefixes are legal, the permutations that *can be read* from σ .

When σ has no d -trigger the condition in (a) holds vacuously, so a trigger read from such a σ always merges. Each move determines S' from (σ, S) and x , and [lemma 2.5\(a\)](#) below shows that $(\sigma x, S')$ is again a scan state, so the stack $\mathcal{S}(\sigma)$ of a legal word is determined by σ . Note that if σ has a d -trigger, then every unread letter $x > q$ is a d -trigger of σx exceeding q , since replacing the last entry of an increasing d -subsequence ending at q by x gives one ending at x . For $d = 1$ a legal move is a new left-to-right minimum, which merges ([figure 1\(a\)](#)), or a letter of the active head, which splits ([figure 1\(b\)](#)). [Figure 2](#) shows the merger of the running example in detail.

Lemma 2.5 (Legal moves). *Let (σ, S) be a scan state and let x be an unread letter.*

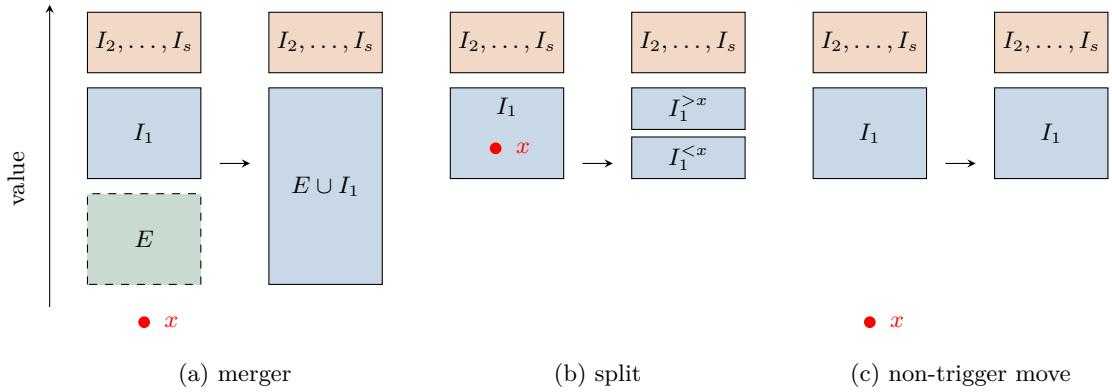


FIGURE 1. The three legal moves of [definition 2.4](#), with values increasing upwards and the active head I_1 at the bottom of the stack. In a merger the letter x lies below the least d -trigger q , and the unread values E strictly between x and q join the head. In a split x lies in the head and cuts it into the values below and above x . A non-trigger move leaves the stack unchanged. The deferred tail is unchanged in all three moves.

- (a) If reading x is a legal move, then $(\sigma x, S')$, with S' as in [definition 2.4](#), is a scan state.
- (b) Reading x is illegal if and only if x lies in one of the deferred intervals $I_2, \dots, I_s$.

Proof. (a) In a merger, x becomes the least d -trigger of σx . If σ has no d -trigger, then S is empty and E is the set of all unread letters above x , so $S' = (E)$ is a scan state with σx . Otherwise the unread letters of σx larger than x are those of E together with those larger than q , that is, $E \cup I_1 \cup \dots \cup I_s$, the union of S' . Every letter of E is smaller than q and every letter of S is larger, so $E < I_1$ and the sets of S' are disjoint and increasing. In a split, the least d -trigger is still q , since $x > q$, and the unread letters above q lose exactly x , which is what the split removes. The two parts of I_1 lie below and above x . In a non-trigger move, $x < q$ if σ has a d -trigger, since an unread $x > q$ is a d -trigger of σx as noted above, so the least d -trigger and the unread letters above it are unchanged, and so is S . If σ has no d -trigger, then S is empty and a non-trigger move leaves it empty.

(b) Let x lie in a deferred interval. Then $x > q$, so x is a d -trigger of σx exceeding q , as noted above. Therefore cases (a) and (c) of [definition 2.4](#) do not apply, and $x \notin I_1$ excludes case (b). Conversely, let reading x be illegal. Since case (c) does not apply, x is a d -trigger of σx . Since case (a) does not apply, x exceeds some d -trigger of σ , so $x > q$. Therefore x lies in the stack, and in a deferred interval since case (b) does not apply. $\square$

Definition 2.6 (Faithful stacks). Let (σ, S) be a scan state with $S = (I_1, \dots, I_s)$. We say that S is *faithful to* σ if a completion w of σ satisfies the residual obligations of all d -triggers of σ precisely when

$$w|_{I_1 \cup \dots \cup I_s} \in \text{Av}(231)(I_1) \oplus \dots \oplus \text{Av}(231)(I_s),$$

where the right-hand side consists of the empty word when $s = 0$.

A scan state fixes the shape of the stack. Faithfulness is the semantic property that the stack describes exactly the obligations created so far, and a stack may or may not have it. [Lemma 2.8](#) shows that a merger preserves faithfulness, and [proposition 2.10](#) that the stack $\mathcal{S}(\sigma)$ of a legal prefix is faithful to σ .

Example 2.7 (Faithful and unfaithful stacks). Let $d = 1$ and let $\sigma = 9, 11, 10, 14$ be the prefix of the running example (4) scanned in [example 2.3](#). Its least trigger is 9, and the unread letters above 9 are 12, 13, 15, so the stacks $\{12, 13\} \mid \{15\}$, $\{12, 13, 15\}$ and $\{12\} \mid \{13\} \mid \{15\}$ all form scan states with σ . For $d = 1$ every letter of σ is a trigger, and a completion w satisfies their residual obligations exactly when 11, 10, 14, $w|_{\{y:y>9\}}$ and 14, $w|_{\{y:y>11\}}$ avoid 231, the obligations of 10 and 14 adding nothing further. As computed in [example 2.3](#), this holds exactly when w reads 12 and 13, in either order, before 15: a completion reading 15 first completes the occurrence 14, 15, y of 231 with $y \in \{12, 13\}$. Therefore $\{12, 13\} \mid \{15\}$, the stack produced by the legal moves, is faithful to σ . The stack $\{12, 13, 15\}$ is not: it allows the order 15, 12, 13, which violates the obligation of 9. Nor is $\{12\} \mid \{13\} \mid \{15\}$: it excludes the order 13, 12, 15, which satisfies every obligation.

Lemma 2.8 (Stack merger). Let (σ, S) be a scan state whose stack $S = (I_1, \dots, I_s)$ is faithful to σ , and let x be an unread letter that is a d -trigger of σx and is smaller than every d -trigger of σ .

- (a) If σ has no d -trigger, then S is empty, and the stack (E) , where E is the set of unread letters larger than x , is faithful to σx . If x is the largest unread letter, then E is empty and the stack is therefore empty.
- (b) If σ has a d -trigger, let q be the least one and let E be the set of unread letters strictly between x and q . Then the merged stack

$$(E \cup I_1, I_2, \dots, I_s),$$

or (E) when $s = 0$, with an empty set deleted, is faithful to σx . In particular only the active head changes.

Proof. Let S' be the new stack, (E) in case (a) and the merged stack in case (b), and let U be the union of its sets. In both cases U is the set of unread letters larger than x , and by lemma 2.5(a) the pair $(\sigma x, S')$ is a scan state.

Let w be a completion of σx . Since x is smaller than every d -trigger of σ , it drops out of every projection created by such a trigger, so w satisfies the residual obligations of the d -triggers of σ exactly when the completion xw of σ does. By faithfulness of S , and since x lies in no I_j , this holds exactly when $w|_{I_1 \cup \dots \cup I_s}$ lies in $\text{Av}(231)(I_1) \oplus \dots \oplus \text{Av}(231)(I_s)$. The one further obligation of σx is that of x , which requires $w|_U$ to avoid 231. We must show that both conditions hold exactly when $w|_U$ lies in the direct sum of S' .

Suppose $w|_U$ lies in the direct sum of S' . A direct sum of 231-avoiding words avoids 231, since positions and values both increase from each block to the next, so $w|_U$ avoids 231. Its restriction to $I_1 \cup \dots \cup I_s$ is the direct sum of the restrictions to $I_1, \dots, I_s$, each of which avoids 231 as a subword of a block, so it lies in the old direct sum.

Conversely, suppose $w|_U$ avoids 231 and $w|_{I_1 \cup \dots \cup I_s}$ lies in the old direct sum. Then $w|_{E \cup I_1}$ avoids 231 as a subword of $w|_U$, each $w|_{I_j}$ avoids 231, and every value of I_1 precedes every value of $I_2 \cup \dots \cup I_s$. Finally, every $e \in E$ precedes every $z \in I_j$ with $j \geq 2$. Suppose instead that z precedes e , and choose $u \in I_1$, which precedes z . Then $e < u < z$, so u, z, e is an occurrence of 231 in $w|_U$, a contradiction. $\square$

The merger creates no separation between E and I_1 , although read letters lie between them in value: the old least trigger q and the earlier triggers above q , some of them read from earlier active heads. When $d \geq 2$, earlier letters that were not triggers may lie inside the value range of E as well, and they create no separation either. All of these letters were read before the new trigger, so none belongs to its projection. No earlier obligation involves E , since every earlier trigger is at least q . A separation produced by a split is therefore not permanent: once the interval below it

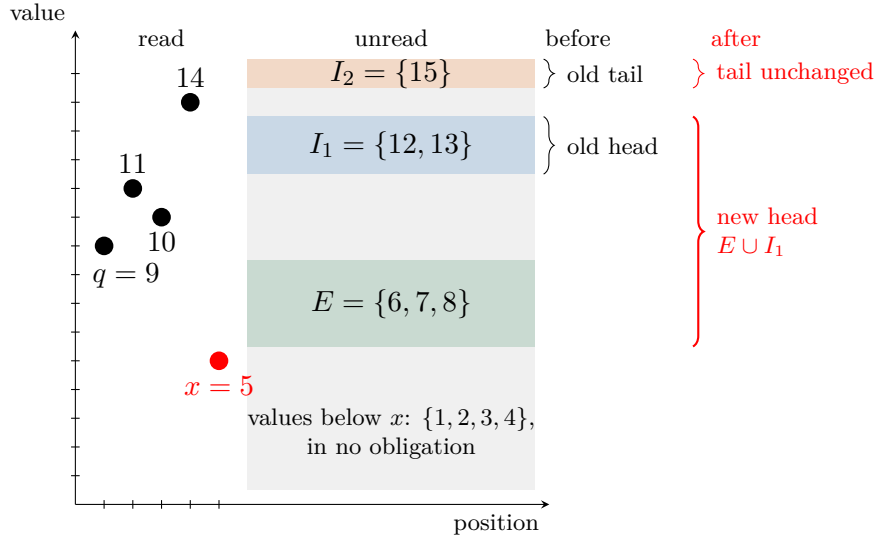


FIGURE 2. The stack merger of lemma 2.8 for $d = 1$ on the running example (4): the prefix 9, 11, 10, 14 has been read, leaving the head $I_1 = \{12, 13\}$ and the tail $I_2 = \{15\}$, and the new trigger $x = 5$ merges $E = \{6, 7, 8\}$, the unread values between x and $q = 9$, into the head. The tail is unchanged. See example 2.9.

has been exhausted, a later merger can extend the head past the letters that produced it, as rows 2–5 of [example 2.12](#) show.

Example 2.9 (One merger). We continue [example 2.3](#). The fifth letter 5 of π is a trigger smaller than the earlier triggers 9, 11, 10, 14, and [figure 2](#) shows the stack before and after reading it. In the notation of [lemma 2.8\(b\)](#), $q = 9$ and $E = \{6, 7, 8\}$, the unread values between 5 and 9. The current stack $\{12, 13\} \mid \{15\}$ consists exactly of the unread values above 9, and it is faithful to the prefix by [example 2.7](#). The merged stack is $\{6, 7, 8, 12, 13\} \mid \{15\}$. The values 6, 7, 8 join the head without a separation: the read letters 9, 10, 11, which lie between them and 12 in value, were read before the trigger 5 and constrain nothing in E . They must, however, precede 15: if 15 came before 6, then either the old obligation is already violated, or 12 precedes 15 and 12, 15, 6 is an occurrence of 231 in the projection of 5, that is, 5, 12, 15, 6 is an occurrence of 1342. Inside the head only 231-avoidance remains. The permutation π continues with 12 before 6, and 13, 12, 6, 7, 8 would have been allowed as well. For $d = 2$ the same permutation is scanned in [example 4.2](#): there the first 2-trigger is 11 rather than 9, and the corresponding merger occurs at the 2-trigger 6, with $q = 10$ and $E = \{7, 8\}$.

Proposition 2.10 (Scan states of avoiders). *Let σ be a legal word with stack $\mathcal{S}(\sigma) = (I_1, \dots, I_s)$.*

- (a) *The stack $\mathcal{S}(\sigma)$ is faithful to σ .*
- (b) *If reading the unread letter x from σ is illegal, then no β_d -avoiding permutation has the prefix σx .*
- (c) *$\mathcal{R}(\sigma)$ is the set of β_d -avoiding permutations of $[n]$ with prefix σ . In particular $\mathcal{R}(\varepsilon) = \text{Av}_n(\beta_d)$, and a word is legal if and only if it is a prefix of a β_d -avoiding permutation.*

Proof. We prove (a) by induction along σ . For the empty word there is no trigger, no obligation and no stack, so the empty stack is faithful. Let $S = \mathcal{S}(\sigma)$ be faithful to σ , let reading x be a legal move, and consider the three cases of [definition 2.4](#).

In a merger, the new stack is that of [lemma 2.8](#), part (a) if σ has no d -trigger and part (b) otherwise, so it is faithful to σx .

In a split, $x \in I_1$, so $x > q$ and x is a d -trigger of σx as noted after [definition 2.4](#). A completion w of σx satisfies the residual obligations of the triggers of σ if and only if the completion xw of σ does, that is, by the induction hypothesis, if and only if $(xw)|_{I_1} \in \text{Av}(231)(I_1)$, $w|_{I_j} \in \text{Av}(231)(I_j)$ for $j \geq 2$, and w reads the sets $I_1 \setminus \{x\}, I_2, \dots, I_s$ in this order. By [lemma 2.2](#), the first condition holds if and only if $w|_{I_1 \setminus \{x\}} \in \text{Av}(231)(I_1^{<x}) \oplus \text{Av}(231)(I_1^{>x})$. Together, these conditions say that the restriction of w to the stack values lies in the direct sum of the split stack. The obligation of the new trigger x adds nothing: its projection is $I_1^{>x} \cup I_2 \cup \dots \cup I_s$, on which the split stack already prescribes a direct sum of 231-avoiding words, and such a word avoids 231 as in the proof of [lemma 2.8](#).

In a non-trigger move, x is not a d -trigger of σx . If σ has no d -trigger, there is no obligation and no stack before or after the move. Otherwise $x < q$, as in the proof of [lemma 2.5](#), so x lies below every d -trigger of σ and belongs to none of their projections. The residual obligations and the stack are unchanged.

For (b), let reading x be illegal. By [lemma 2.5\(b\)](#), x lies in a deferred interval I_j , $j \geq 2$. Every completion of σ beginning with x reads x before the letters of the nonempty set I_1 , so its restriction to the stack values is not in the direct sum of [definition 2.6](#), and by (a) it violates the residual obligation of some d -trigger c of σ : the projection created by c contains 231, and the permutation contains β_d by [lemma 2.1](#).

For (c), let π be a permutation with prefix σ . If π avoids β_d , then every prefix of π is legal: the empty prefix is, and if a prefix σ' of π is legal and the next letter x of π were illegal, (b) would forbid the prefix $\sigma'x$ of the avoider π . Therefore $\pi \in \mathcal{R}(\sigma)$. Conversely, let $\pi \in \mathcal{R}(\sigma)$. Then π itself is legal, no letter is unread, the stack is empty, and by (a) the empty completion satisfies

the residual obligation of every d -trigger of π . That is, the projection of every d -trigger avoids 231 after standardization, and π avoids β_d by [lemma 2.1](#). With $\sigma = \varepsilon$ this gives $\mathcal{R}(\varepsilon) = \text{Av}_n(\beta_d)$, and a prefix of an avoider is legal by the first paragraph of this part. Conversely, a legal σ has an avoiding completion: read the unread letters smaller than q , or all unread letters when σ has no d -trigger, in decreasing order, and then the intervals $I_1, \dots, I_s$ in order, each in decreasing order. Each letter of the first group is the largest unread letter below the least trigger, or the largest unread letter when there is none, so it is a non-trigger move or a merger with $E = \emptyset$, and the stack is unchanged. Each later letter is the greatest element of the active head, and therefore a d -trigger above q , so reading it is an endpoint choice. All moves are legal, so the completed permutation lies in $\mathcal{R}(\sigma)$ and avoids β_d . $\square$

Being a prefix of a β_d -avoiding permutation is stronger than avoiding β_d . The word 134 avoids 1342, but its only completion in [4] is 1342, and after 1, 3 the stack is $\{2\} \mid \{4\}$, so reading 4 is illegal.

2.2. The one-threshold state. For $d = 1$, every letter is a trigger, so the least d -trigger q of [definition 2.4](#) is the least letter m read so far. Before the first letter, m is a virtual value larger than every letter. A new left-to-right minimum joins the unread values between it and m to the active head, or creates the stack when the stack is empty. A letter of the active head shortens the head at an endpoint choice and replaces it by its lower and upper subintervals at an interior choice. By [definition 2.4](#), the stack of a legal prefix consists of the unread letters above m , and by [proposition 2.10\(c\)](#) the legal prefixes are the prefixes of 1342-avoiding permutations. Two unread values are *adjacent* if no unread value lies strictly between them. The following lemma determines the separations of the stack.

Lemma 2.11 (Separations of the one-threshold stack). *After a legal prefix has been read, let $u < v$ be adjacent unread letters above m . Then u and v lie in different intervals of the stack if and only if some letter x with $u < x < v$ was read after a letter smaller than u , that is, at a time when the minimum read so far was already below u .*

Proof. We argue by induction along the scan. For the empty prefix there is nothing to prove. The next legal move reads a new minimum or a letter of the active head.

Suppose first that the next letter x is a new minimum, joining E to the head or creating the stack (E). An adjacent pair $u < v$ above the new minimum that was not an adjacent pair above the old minimum has $u \in E$. Every letter read before x exceeds the old minimum and hence u , so no letter between u and v was read after a letter smaller than u . The value v lies in E or is the least element of I_1 , and $E \cup I_1$, or E when the old stack was empty, is a single interval of the new stack. Thus neither condition holds for such a pair.

Suppose instead that x is read from the active head I_1 . Only the unread neighbours $u < x < v$ of x can form a new adjacent pair, and only when both exist and lie above m . This excludes $x = \min I_1$, where u lies below m . For an interior choice, u and v lie in the two parts created by the split. For $x = \max I_1$, u lies in the shortened head and v is the least element of I_2 if a deferred interval remains. Otherwise x is the largest unread letter and no new pair arises. In every case u and v lie in different intervals, and x , read after $m < u$, witnesses the second condition.

In both cases every other adjacent pair keeps its members, its intervals, and the letters between them, so both conditions are unchanged for it. $\square$

Therefore the stack is recovered from the scanned prefix alone: its values are the unread letters above m , and its intervals are the blocks into which this increasing list is cut at exactly the adjacent pairs of the lemma. The precedence recorded by a separation is also visible directly: if x was read after $c < u$ and v preceded u , then c, x, v, u would form 1342. Conversely, two letters of the same interval can occur in either order in a 1342-avoiding completion. [Corollary 4.4](#) proves

this and extends both statements to every d . Adjacency cannot be dropped: after the prefix 3, 5, 1 of a 1342-avoiding permutation of $[7]$ the stack is $\{2, 4\} \mid \{6, 7\}$, so 2 and 6 lie in different intervals, although no letter between them was read after a letter smaller than 2.

Example 2.12 (A one-threshold scan). We scan the whole permutation π of (4), which can be read down the second column of the table below. The table lists, after each letter π_i , the type of move, the current minimum m , the interval stack, and the symbol $W_p(L)$ of section 2.3, where p is the number of unread letters below m , L lists the interval sizes, and $W_p(L)$ is the number of 1342-avoiding completions of the prefix. Every new minimum is a merger in the sense of definition 2.4(a). The table writes “merger” when it joins values to the head or creates the stack and “new minimum” when $E = \emptyset$, and it writes “head exhausted” for an endpoint choice in a head of size one.

i	π_i	move	m	interval stack	$W_p(L)$
1	9	new minimum, merger	9	$\{10, \dots, 15\}$	$W_8((6))$
2	11	interior choice	9	$\{10\} \mid \{12, \dots, 15\}$	$W_8((1, 4))$
3	10	head exhausted	9	$\{12, \dots, 15\}$	$W_8((4))$
4	14	interior choice	9	$\{12, 13\} \mid \{15\}$	$W_8((2, 1))$
5	5	new minimum, merger	5	$\{6, 7, 8, 12, 13\} \mid \{15\}$	$W_4((5, 1))$
6	12	interior choice	5	$\{6, 7, 8\} \mid \{13\} \mid \{15\}$	$W_4((3, 1, 1))$
7	6	endpoint choice	5	$\{7, 8\} \mid \{13\} \mid \{15\}$	$W_4((2, 1, 1))$
8	2	new minimum, merger	2	$\{3, 4, 7, 8\} \mid \{13\} \mid \{15\}$	$W_1((4, 1, 1))$
9	3	endpoint choice	2	$\{4, 7, 8\} \mid \{13\} \mid \{15\}$	$W_1((3, 1, 1))$
10	8	endpoint choice	2	$\{4, 7\} \mid \{13\} \mid \{15\}$	$W_1((2, 1, 1))$
11	4	endpoint choice	2	$\{7\} \mid \{13\} \mid \{15\}$	$W_1((1, 1, 1))$
12	7	head exhausted	2	$\{13\} \mid \{15\}$	$W_1((1, 1))$
13	1	new minimum	1	$\{13\} \mid \{15\}$	$W_0((1, 1))$
14	13	head exhausted	1	$\{15\}$	$W_0((1))$
15	15	head exhausted	1	$\emptyset$	$W_0(\emptyset)$

The first five rows are examples 2.3 and 2.9. In the terms of lemma 2.11, the letters 9, 10, and 11 separate nothing at row 5, as in example 2.9, while 14, read after $9 < 13$, separates 13 from 15 from row 4 on. The interior choice 12 then creates three intervals, separated between 8 and 13 and between 13 and 15. After the endpoint choice 6, the new minimum 2 joins $\{3, 4\}$ to the head. The head $\{3, 4, 7, 8\}$ contains the read letters 5 and 6 in its range, again without a separation. At row 8 the unread letters above $m = 2$ are 3, 4, 7, 8, 13, 15, and the lemma cuts this list between 8 and 13, with witness 12 read after $5 < 8$, and between 13 and 15, with witness 14, and nowhere else. This is the stack of the table. The head is then consumed in the 231-avoiding order 3, 8, 4, 7. The new minimum 1 joins nothing to the head, and 13, 15 complete the scan. Throughout, the later minima 5, 2, and 1 are read between letters of the stack, and only sizes enter the state: the heads $\{6, 7, 8, 12, 13\}$ of row 5 and $\{3, 4, 7, 8\}$ of row 8 contribute to L only their sizes 5 and 4.

2.3. The one-threshold recurrence. Recall that m is the least letter in the scanned prefix, virtual for the empty prefix. Let p be the number of unread letters below m , which we call the *control* of the state, and let $L = (\ell_1, \dots, \ell_s)$ be the positive sizes of the active and deferred intervals. For a legal prefix with these data, let $W_p(L)$ be the number of completions that produce a 1342-avoiding permutation. By proposition 2.10(c), these completions are exactly the sequences of legal moves that read every unread letter, and by definition 2.4 the legal moves from a legal prefix, their successor states and their multiplicities depend only on (p, L) and on the local ranks of the letters read. Therefore this number depends only on (p, L) . We write $|L| = \sum_{i=1}^s \ell_i$ and $\text{len}(L) = s$. If L' is a list, $(\ell)L'$ and $(a, b)L'$ denote concatenation with the displayed initial entries. In the equations below, each summand stands for reading one letter and moving from the current

state to the state indexed by the summand. We call such a move a *transition*. A summand with coefficient 2 stands for two transitions with the same successor, which we label by the local rank of the letter read.

Proposition 2.13 (One-threshold recurrence). *For $L = (\ell)L'$ with $\ell \geq 1$,*

$$\begin{aligned} W_p((\ell)L') &= \sum_{h=0}^{p-1} W_h((\ell + p - 1 - h)L') + V_\ell(p, L') \\ &\quad + \sum_{\substack{a, b \geq 1 \\ a+b=\ell-1}} W_p((a, b)L'), \end{aligned} \quad (5)$$

where

$$V_\ell(p, L') = \begin{cases} W_p(L'), & \ell = 1, \\ 2W_p((\ell - 1)L'), & \ell \geq 2. \end{cases} \quad (6)$$

The boundary and initial conditions are

$$W_p(\emptyset) = \begin{cases} 1, & p = 0, \\ W_{p-1}(\emptyset) + \sum_{h=0}^{p-2} W_h((p - 1 - h)), & p \geq 1, \end{cases} \quad (7)$$

$$|\text{Av}_n(1342)| = W_n(\emptyset). \quad (8)$$

These equations count every 1342-avoiding permutation exactly once, as a maximal sequence of transitions from $(n, \emptyset)$.

Proof. The avoiding completions are the sequences of legal moves that read every unread letter, and the legal moves are the new minima and the letters of the active head. We partition the possible next letters by their location. If the next letter is below m , let h be the number of unread letters below it. There is one letter for each $0 \leq h < p$. The new minimum leaves h unrestricted lower letters, while the $p - 1 - h$ letters between it and m join the active head. This gives the first sum in (5). If the active head has size one, it has one endpoint. If its size is at least two, its two endpoints have the same standardized successor. These choices give the endpoint term $V_\ell(p, L')$ of (6), the coefficient 2 standing for the two transitions of local ranks 1 and ℓ . An interior letter replaces the active head by two ordered subintervals of sizes $a, b \geq 1$ with $a + b = \ell - 1$, so that L becomes $(a, b)L'$, giving the last sum in (5) by lemma 2.2. With an empty stack only a new minimum can be read, which gives (7).

The quantity

$$\rho_1(p, L) = p + |L| \quad (9)$$

decreases by one under every transition, so every maximal transition sequence ends at $(0, \emptyset)$. In the other direction, the selected local rank at each step recovers a unique next letter in the current ordered alphabet. Maximal transition sequences and avoiding completions are therefore inverse constructions, so the equations count every avoiding completion exactly once. For the empty prefix, $p = n$ and $L = \emptyset$, which is (8). $\square$

Example 2.14 (All one-threshold moves). After the prefix 9, 11, 10, 14, 5, 12 of the running example (4), row 6 of the table in example 2.12, the unread letters below the current minimum 5 are 1, 2, 3, 4 and the stack is $\{6, 7, 8\} \mid \{13\} \mid \{15\}$. The state is $(p, L) = (4, (3, 1, 1))$, and (5) gives

$$\begin{aligned} W_4((3, 1, 1)) &= W_0((6, 1, 1)) + W_1((5, 1, 1)) + W_2((4, 1, 1)) + W_3((3, 1, 1)) \\ &\quad + 2W_4((2, 1, 1)) + W_4((1, 1, 1, 1)). \end{aligned}$$

The first four terms correspond to reading 1, 2, 3, or 4, which join $\{2, 3, 4\}$, $\{3, 4\}$, $\{4\}$, or nothing to the active head. The fifth term records the two endpoints 6, 8 of the active head, and the last

records its interior letter 7. Neither 13 nor 15 can be read legally until the active head is empty. The scan of [example 2.12](#) continues with the endpoint 6.

Example 2.15 (Counting $\text{Av}_5(1342)$ by the literal recurrence). By (8) and (7),

$$|\text{Av}_5(1342)| = W_5(\emptyset) = W_0((4)) + W_1((3)) + W_2((2)) + W_3((1)) + W_4(\emptyset),$$

the five terms corresponding to the first letter 1, 2, 3, 4, 5. Expanding by (5) and (6) reaches twenty distinct states in all, listed here with their values, in increasing order of $\rho_1 = p + |L|$:

ρ_1	values
0	$W_0(\emptyset) = 1$
1	$W_0((1)) = 1, W_1(\emptyset) = 1$
2	$W_0((1, 1)) = 1, W_0((2)) = 2, W_1((1)) = 2, W_2(\emptyset) = 2$
3	$W_0((1, 2)) = 2, W_0((2, 1)) = 2, W_0((3)) = 5, W_1((1, 1)) = 3, W_1((2)) = 6, W_2((1)) = 6, W_3(\emptyset) = 6$
4	$W_0((4)) = 14, W_1((3)) = 20, W_2((2)) = 23, W_3((1)) = 23, W_4(\emptyset) = 23$
5	$W_5(\emptyset) = 103$

For example,

$$W_0((4)) = 2W_0((3)) + W_0((1, 2)) + W_0((2, 1)) = 10 + 2 + 2 = 14,$$

$$W_1((3)) = W_0((3)) + 2W_1((2)) + W_1((1, 1)) = 5 + 12 + 3 = 20,$$

$$W_2((2)) = W_0((3)) + W_1((2)) + 2W_2((1)) = 5 + 6 + 12 = 23,$$

$$W_3((1)) = W_0((3)) + W_1((2)) + W_2((1)) + W_3(\emptyset) = 5 + 6 + 6 + 6 = 23,$$

so that $W_5(\emptyset) = 14 + 20 + 23 + 23 + 23 = 103$. In the first line the terms are the two endpoints and the interior choices $(a, b) = (1, 2), (2, 1)$, and in the second the new minimum, which joins nothing to the head, the two endpoints, and the interior choice $(1, 1)$. The states with $p = 0$ are counted by the Catalan numbers $C_\ell = \frac{1}{\ell+1} \binom{2\ell}{\ell}$, that is, $W_0((\ell)) = C_\ell$, as [lemma 2.2](#) predicts. The number of distinct states grows exponentially with n ([proposition 2.16](#) below). [Example 2.21](#) recomputes the same value from the kernel rows of the shrunk state space.

2.4. The exponential literal state space. The recurrence (5) is finite for each n , but memoizing its composition-valued argument does not give a polynomial bound.

Proposition 2.16. *For $n \geq 2$, the computation of $W_n(\emptyset)$ reaches at least F_{n-1} distinct composition arguments, where F_{n-1} is the $(n-1)$ st Fibonacci number, $F_1 = F_2 = 1$.*

Proof. Choosing the letter 1 at the initial state reaches $W_0((n-1))$. Repeated interior choices reach every positive composition L satisfying

$$|L| + \text{len}(L) = n. \tag{10}$$

Reversing an interior split replaces $(a, b, \dots)$ by $(a + b + 1, \dots)$, and repeated reversals return every such composition to $(n-1)$. If L has s parts, then $|L| = n - s$, so the number of choices is $\binom{n-s-1}{s-1}$. Summing over $1 \leq s \leq \lfloor n/2 \rfloor$ gives

$$\sum_{s=1}^{\lfloor n/2 \rfloor} \binom{n-s-1}{s-1} = F_{n-1}. \quad \square$$

Literal memoization of (5) therefore has exponentially many states. The values of the ordered parts cannot be replaced by their total and their number. For example, when $p = 0$, (3) gives $W_0((1, 3)) = C_1 C_3 = 5$ and $W_0((2, 2)) = C_2^2 = 4$. The next construction removes the entire composition from the memoization key while retaining its effect on completion counts.

2.5. Protected-tail factorization. Before the general statement, consider the smallest nontrivial instance of (5): one unread letter below the current minimum and a head of size one, with an arbitrary tail L' . Either the head letter is read first, which leaves the stack L' at control 1, or the lower letter is read first and the head letter next, which leaves L' at control 0. Therefore

$$W_1((1)L') = W_1(L') + W_0(L').$$

The two coefficients count the ways to reach the stack L' at controls 1 and 0, and they do not depend on L' . Writing them as $K_1(1, 1) = K_1(1, 0) = 1$, the identity reads $W_1((1)L') = \sum_{t \in \{0,1\}} K_1(1, t) W_t(L')$. The theorem below shows when such a factorization holds for a recurrence on a control set and a stack: the transitions before the tail is exposed must leave the tail unchanged and must not depend on it. We state it for an abstract recurrence because we apply it twice, to (5) below and to the recurrence for general d in section 5.

Let $\mathcal{C}$ be a set and $\mathcal{B}^*$ the set of finite words over an alphabet $\mathcal{B}$. In the application below, an element of $\mathcal{C}$ records the number of unread values that the obligations so far leave unconstrained, and a word in $\mathcal{B}^*$ lists the sizes of the pending intervals, active head first. A *recurrence* H on $\mathcal{C} \times \mathcal{B}^*$ assigns to each nonboundary state (c, Y) an equation

$$H_c(Y) = \sum_e H_{c_e}(Y_e)$$

with finitely many summands, a summand with coefficient m being written as m equal summands, as in section 2.3. A boundary state has a prescribed nonnegative-integer value and no equation. Each summand is a transition $(c, Y) \rightarrow (c_e, Y_e)$, so parallel transitions may occur. A *path* is a finite sequence of consecutive transitions, and a *complete path* is one ending at a boundary state. The recurrence *terminates* if it has no infinite path. Then every state has finitely many complete paths, and induction along the well-founded transition relation shows that $H_c(Y)$ is the sum, over the complete paths from (c, Y) , of the boundary values at their last states.

For a factorization $Y = UL$ of a stack we write $U \mid L$ when the suffix L is the one under consideration. The marker is a notational device and not part of the state. In the application, L is a tail of the interval stack at a separation, chosen freely. A state is an *exposure* of L if its stack is exactly L , that is, if it has the form $(t, \emptyset \mid L)$. Given a state $s_0 = (c, U \mid L)$ with $U \neq \emptyset$, we call L *protected before exposure from s_0* if the following two conditions hold.

- (P1) Every state reachable from s_0 by a path containing no exposure of L has the form $(c', U' \mid L)$ with $U' \neq \emptyset$.
- (P2) For every such state and every word $L^* \in \mathcal{B}^*$, replacing L by L^* induces a bijection between the transitions leaving $(c', U' \mid L)$ and those leaving $(c', U' \mid L^*)$, with corresponding targets $(c'', V' \mid L)$ and $(c'', V' \mid L^*)$.

Thus (P1) says that the suffix L survives untouched until it is exposed, and (P2) says that the transitions before exposure do not depend on it. We call L *necessarily exposed from s_0* if every complete path from s_0 contains an exposure of L , possibly as its last state.

Theorem 2.17 (Protected-tail factorization). *Let H be a terminating recurrence on $\mathcal{C} \times \mathcal{B}^*$. Suppose that, for every $c \in \mathcal{C}$, $x \in \mathcal{B}$, and $L \in \mathcal{B}^*$, the marked suffix L is protected before exposure from $(c, x \mid L)$ and necessarily exposed from that state.*

Let $K_x^L(c, t)$ be the number of paths from $(c, x \mid L)$ to $(t, \emptyset \mid L)$ whose only exposure of L is their last state. We call these paths stopped at the first exposure of L . Then $K_x^L(c, t)$ is independent of L . We write $K_x(c, t)$ for it. If $Y = x_1 \cdots x_s$ and $G_t = H_t(\emptyset)$, then

$$H_c(Y) = \sum_{t \in \mathcal{C}} (K_{x_1} K_{x_2} \cdots K_{x_s})(c, t) G_t. \quad (11)$$

Here

$$(AB)(c, t) = \sum_{u \in \mathcal{C}} A(c, u)B(u, t),$$

and the empty product is the identity kernel $\mathbf{1}_{c=t}$, which is 1 if $c = t$ and 0 otherwise. All displayed sums are finite by termination.

Proof. By (P1) and (P2) and induction on the number of transitions, replacing L by L^* gives a bijection between the paths from $(c, x \mid L)$ that contain no exposure of L and the paths from $(c, x \mid L^*)$ that contain no exposure of L^* , with equal controls at corresponding states. The transition that produces the first exposure leaves a state of the form $(c', U' \mid L)$ and is matched as well. Therefore $K_x^L(c, t)$ is independent of L . By necessary exposure, every complete path from $(c, x \mid L)$ has a unique first exposure of L , of the form $(t, \emptyset \mid L)$. Cutting the path there gives a stopped path counted by $K_x(c, t)$ followed by a complete path from (t, L) , and concatenation is the inverse operation. Summing the boundary values over the complete paths from $(c, x \mid L)$ therefore gives

$$H_c(xL) = \sum_{t \in \mathcal{C}} K_x(c, t)H_t(L).$$

Applying this identity successively to $x_1, \dots, x_s$ proves (11). $\square$

Remark 2.18 (Pushdown systems and pop sequences). Conditions (P1) and (P2) say that the recurrence is a pushdown system: a transition depends only on the control and on the part of the stack before the marker, and the part after it is untouched until exposed, which a pushdown automaton satisfies by definition. The kernel $K_x(c, t)$ counts the runs that pop x while passing from c to t , and [theorem 2.17](#) is the counting form of the classical conversion of a pushdown automaton into a context-free grammar, whose variable $[cx t]$ describes exactly those runs, see Hopcroft, Motwani, and Ullman [17, Theorem 6.14] or Sipser [23, Lemma 2.27]. The same runs are the pop sequences of the weighted pushdown systems of Reps, Schwoon, Jha, and Melski [22], and (11) is the first-passage decomposition of weighted lattice paths, see Flajolet and Sedgewick [14, Sections I.5.3 and V.4]. We state and prove the theorem because the control set and the alphabet are infinite in our application, so the textbook statements do not apply verbatim. The consequence also differs: with infinitely many controls the construction gives an infinite grammar, so the Chomsky–Schützenberger theorem [11] does not apply and no algebraicity follows, and what the factorization provides is a memoization key of polynomial size ([theorem 1.1](#)) rather than a generating function.

2.6. The scalar kernel algorithm. We apply [theorem 2.17](#) to the recurrence (5), with $\mathcal{C} = \mathbb{N}$ the set of values of p , with $\mathcal{B}$ the set of positive integers, and with W in the role of the recurrence H . A word in $\mathcal{B}^*$ is then the list L of interval sizes. We place the marker before an arbitrary tail L of the stack. A new minimum can enlarge the active head, an endpoint can shorten it, and an interior letter can replace it by two new parts. None of these moves changes L before every part preceding the marker has disappeared. Equations (5) and (6) show that, before exposure, the transitions leaving a state and their targets are determined by p and the blocks preceding the marker. Replacing L therefore gives the bijection of outgoing transitions required by (P2). Every move decreases (9), and the only terminal state has an empty stack. Therefore no complete path can terminate before exposing L . Thus [theorem 2.17](#) applies.

For $\ell \geq 1$, let $K_\ell(p, t)$ be the kernel of [theorem 2.17](#), the number of stopped paths from $(p, (\ell) \mid L)$ to $(t, \emptyset \mid L)$, which does not depend on L . Recall that the control p is the number of unread letters below the current minimum. In the kernel we call p the *source control*, t a *terminal control*, and $K_\ell(p, \cdot)$ a *scalar kernel row*. A transition that selects an unread letter below the current minimum is a *base move*. A transition that selects a value from the active head is an

active-head move. Then

$$W_p((\ell)L) = \sum_{t \geq 0} K_\ell(p, t) W_t(L). \quad (12)$$

Put $K_0(p, t) = \mathbf{1}_{p=t}$ and $G_p = W_p(\emptyset)$.

Proposition 2.19 (Scalar kernel recurrences). *For $\ell \geq 1$,*

$$\begin{aligned} K_\ell(p, t) = & \sum_{h=0}^{p-1} K_{\ell+p-1-h}(h, t) + D_\ell(p, t) \\ & + \sum_{\substack{a, b \geq 1 \\ a+b=\ell-1}} \sum_{u \geq 0} K_a(p, u) K_b(u, t), \end{aligned} \quad (13)$$

where

$$D_1(p, t) = \mathbf{1}_{p=t}, \quad D_\ell(p, t) = 2K_{\ell-1}(p, t) \quad (\ell \geq 2). \quad (14)$$

The empty-stack values satisfy

$$G_p = \begin{cases} 1, & p = 0, \\ \sum_{h=0}^{p-1} \sum_{t \geq 0} K_{p-1-h}(h, t) G_t, & p \geq 1, \end{cases} \quad |\text{Av}_n(1342)| = G_n. \quad (15)$$

Proof. We partition the stopped paths by their first move. A new minimum leaving h lower letters gives the first sum in (13). If $\ell = 1$, consuming the only active letter exposes L at the same control. If $\ell \geq 2$, either endpoint leaves an active head of size $\ell - 1$. These are the terms in (14). An interior letter creates two parts of sizes $a, b \geq 1$ with $a + b = \ell - 1$. We move the marker to $(p, (a) \mid (b)L)$. A path stopped at the first exposure of L passes through a unique first exposure of $(b)L$, because every transition changes the stack only in its current active head. Cutting there splits the path into one counted by $K_a(p, u)$, for some intermediate control u , and one counted by $K_b(u, t)$. Concatenation inverts the cut. Summing over u gives the double sum in (13). Finally, (15) follows from (7): a new minimum leaving h lower letters creates a part of size $p - 1 - h$, to which (12) applies, and K_0 covers the case $h = p - 1$. $\square$

Lemma 2.20 (Exact scalar support). *For every $p \geq 0$ and $\ell \geq 1$, the support of the row $K_\ell(p, \cdot)$, the set of t with $K_\ell(p, t) \neq 0$, is*

$$\text{supp } K_\ell(p, \cdot) = \{0, 1, \dots, p\}. \quad (16)$$

In addition, $K_\ell(p, p) = C_\ell$.

Proof. A base move strictly decreases the control, while an active-head move leaves it unchanged. Thus every terminal control is at most p . A path ending at p uses no base move. By lemma 2.2, such paths are in bijection with the 231-avoiding permutations of the original ℓ -element interval, and hence are counted by C_ℓ .

For $t < p$, consider the path that first reads the base letter with exactly t unread base letters below it, which makes the control t and the active head of size $\ell + p - 1 - t$, and then reads endpoints until the head is exhausted. It reads no other base letter and exposes the marked suffix at control t . Thus every $t \in \{0, \dots, p\}$ occurs. $\square$

We call $p + \ell$ the *grade* of the kernel row $K_\ell(p, \cdot)$.

Example 2.21 (From kernel rows to $|\text{Av}_5(1342)|$). We write each row $K_\ell(p, \cdot)$ as the vector $(K_\ell(p, 0), \dots, K_\ell(p, p))$, which by lemma 2.20 lists its whole support. Equations (13)–(14) give

$$K_1(1, \cdot) = (1, 1), \quad K_3(0, \cdot) = (5), \quad K_2(1, \cdot) = (4, 2), \quad K_1(2, \cdot) = (3, 1, 1),$$

for example $K_1(2, \cdot) = K_2(0, \cdot) + K_1(1, \cdot) + \mathbf{e}_2 = (2) + (1, 1) + \mathbf{e}_2 = (3, 1, 1)$, where $\mathbf{e}_t(u) = \mathbf{1}_{u=t}$. The rows of grade 4 follow:

$$\begin{aligned} K_4(0, \cdot) &= 2K_3(0, \cdot) + K_1(0, 0)K_2(0, \cdot) + K_2(0, 0)K_1(0, \cdot) = (10) + (2) + (2) = (14), \\ K_3(1, \cdot) &= K_3(0, \cdot) + 2K_2(1, \cdot) + \sum_u K_1(1, u)K_1(u, \cdot) = (5) + (8, 4) + (2, 1) = (15, 5), \\ K_2(2, \cdot) &= K_3(0, \cdot) + K_2(1, \cdot) + 2K_1(2, \cdot) = (5) + (4, 2) + (6, 2, 2) = (15, 4, 2), \\ K_1(3, \cdot) &= K_3(0, \cdot) + K_2(1, \cdot) + K_1(2, \cdot) + \mathbf{e}_3 = (12, 3, 1, 1). \end{aligned}$$

Since $(G_0, \dots, G_4) = (1, 1, 2, 6, 23)$, (15) gives

$$\begin{aligned} G_5 &= 14G_0 + (15G_0 + 5G_1) + (15G_0 + 4G_1 + 2G_2) + (12G_0 + 3G_1 + G_2 + G_3) + G_4 \\ &= 14 + 20 + 23 + 23 + 23 = 103, \end{aligned}$$

the five groups corresponding, in order, to $h = 0, 1, 2, 3, 4$, the last through the identity kernel K_0 . The five group sums are the five terms $W_0((4)), \dots, W_4(\emptyset)$ of [example 2.15](#): each kernel row compresses the subtree of the literal recurrence below one first-letter choice into a vector indexed by the control at which the empty tail is exposed.

Theorem 2.22 (Polynomial enumeration of 1342-avoiders). *For every $N \geq 0$, the numbers $|\text{Av}_n(1342)|$ for $0 \leq n \leq N$ can be computed using $O(N^5)$ integer arithmetic operations and $O(N^3)$ stored integers. Every integer has $O(N \log N)$ bits. Therefore the bit time is $O(N^5 M(N \log N))$ and the bit storage is $O(N^4 \log N)$, where $M(b)$ is the cost of multiplying b -bit integers, as in [theorem 1.1](#).*

Proof. We order the kernel rows by grade. The row in a base term of (13) has grade $h + (\ell + p - 1 - h) = p + \ell - 1$. In a split term, $p + a < p + \ell$. If $K_a(p, u) \neq 0$, then $u \leq p$ by [lemma 2.20](#), and hence $u + b \leq p + b < p + \ell$. The term D_1 is initial data, and D_ℓ for $\ell \geq 2$ uses the row of grade $p + \ell - 1$. Thus increasing grade is a topological order for all kernel dependencies: an order in which every row comes after all the rows its recurrence uses, so that evaluating the rows in this order never meets a row not yet computed. Equation (15) is then evaluated in increasing p , because every terminal control t in its nonzero terms satisfies $t \leq h < p$.

We compute every row in the *padded region* $p \geq 0$, $\ell \geq 1$, $p + \ell \leq N$: it is closed under the dependencies in the preceding paragraph and contains every row with $\ell \geq 1$ needed to evaluate (15) for $n \leq N$, together with rows that no such evaluation reaches. It contains $\binom{N+1}{2} = O(N^2)$ source rows. By [lemma 2.20](#), these rows have at most

$$\sum_{\ell=1}^N \sum_{p=0}^{N-\ell} (p+1) = \binom{N+2}{3} = O(N^3)$$

nonzero entries in total. For one source row, a sparse evaluation of all split terms has $O(N)$ choices of a , $O(N)$ possible intermediate controls u , and $O(N)$ terminal controls in the row being added. It therefore uses $O(N^3)$ operations per source row and $O(N^5)$ operations in total. The nonsplit kernel terms use $O(N^4)$ operations in total, and (15) uses $O(N^3)$.

For a scalar kernel entry, we take the marked suffix to be empty, by suffix independence. Its initial number of unread letters is then $p + \ell \leq N$. An empty-stack computation also starts with at most N unread letters. This number decreases at every transition. At each state, the sum of the outgoing transition multiplicities is the number of legal next letters and is at most N . Thus each stored count is at most $(N+1)^N$ and has $O(N \log N)$ bits. Accounting for integer multiplication and for the bit length of every stored entry gives the stated bit bounds. Every product and partial accumulator is a nonnegative contribution to a final kernel or empty-stack value, so the same bound applies to intermediate integers. $\square$

Bóna [9] found the algebraic generating function

$$\sum_{n \geq 0} |\text{Av}_n(1342)|x^n = \frac{(1 - 8x)^{3/2} - 8x^2 + 20x + 1}{2(1 + x)^3}. \quad (17)$$

[Theorem 2.22](#) is a separate transfer construction, and it extends to the family: we replace p by d ordered band counts, while the interval stack, its merger rule, and the protected-tail theorem remain unchanged.

3. FROM ONE THRESHOLD TO MANY

We now consider the pattern β_d for an arbitrary $d \geq 1$, and replace the current minimum of [section 2](#) by d thresholds and bands, which record the base values that a new letter may update. After a prefix has been read, let b_j be the smallest final entry of an increasing subsequence of length j contained in that prefix, for $1 \leq j \leq d$. When the prefix contains no such subsequence, we use the virtual value $b_j = n + j$. Then

$$b_1 < b_2 < \dots < b_d.$$

We call $b_1, \dots, b_d$ the *patience-sorting thresholds* of the prefix, since they are the top cards of the first d piles when the prefix is dealt by patience sorting, see Aldous and Diaconis [3]. Every nonvirtual threshold value has already been read. We call the unread values below b_d the *base values*. They fall into d open bands, shown in [figure 3](#). Band 0 lies below b_1 , while band i lies between b_i and b_{i+1} for $1 \leq i < d$. We write $\mathbf{p} = (p_0, p_1, \dots, p_{d-1})$ for the band sizes. We again call $\mathbf{p}$ the *control*, write $\|\mathbf{p}\|_1 = p_0 + \dots + p_{d-1}$ for its *control mass*, and $\mathbf{0}$ for the zero control. Let $B_0, \dots, B_{d-1}$ be the corresponding unread blocks. Together with the pending intervals, the value order is

$$B_0 < b_1 < B_1 < b_2 < \dots < b_{d-1} < B_{d-1} < b_d < I_1 < \dots < I_s. \quad (18)$$

Empty blocks are allowed. As in the one-threshold case ([section 2](#)), a base move selects a value from some B_i and an active-head move selects a value from I_1 . We call a base move from B_i with $i < d - 1$ an *early-band move* and a base move from B_{d-1} a *last-band move*. In the terms of [definition 2.4](#), a value of B_i with $i < d - 1$ is a non-trigger move, a value of B_{d-1} is a merger, and a value of I_1 is a split ([proposition 4.1\(b\)](#) below). By [lemma 3.1](#) below, the least d -trigger q of [definition 2.4](#) is b_d whenever b_d is nonvirtual. Initially $\mathbf{p} = (n, 0, \dots, 0)$.

If the next base value x lies in band $i \geq 1$, then x ends an increasing subsequence of length $i + 1$, since $b_i < x$. For $i = 0$, the letter x is an increasing subsequence of length one. In either case, x ends no increasing subsequence of length $i + 2$, since that would imply $b_{i+1} < x$, while $x \in B_i$ gives $x < b_{i+1}$. Thus only b_{i+1} changes. A base value creates a new d -trigger exactly when it lies in band $d - 1$. The first $d - 1$ bands update one threshold without creating an obligation. The last band also creates a 231-avoidance obligation.

Lemma 3.1 (Least-trigger threshold). *After a prefix has been read, either b_d is virtual and no d -trigger has been read, or b_d is the smallest value of a d -trigger in the prefix. In particular, no unread value below b_d belongs to the projection of any earlier trigger, so none is constrained by an obligation created so far. When b_d is nonvirtual, every unread value above b_d belongs to the projection created by the trigger b_d .*

Proof. If b_d is virtual, no increasing d -subsequence has occurred, so no d -trigger has been read. Otherwise b_d is by definition the least final value of an increasing d -subsequence in the prefix, and hence is itself a d -trigger, and every other d -trigger has value at least b_d . An unread value $x < b_d$ is therefore smaller than every earlier trigger and occurs in none of their projections. If b_d is nonvirtual and $x > b_d$ is unread, then x occurs after the trigger b_d and belongs to its projection. $\square$

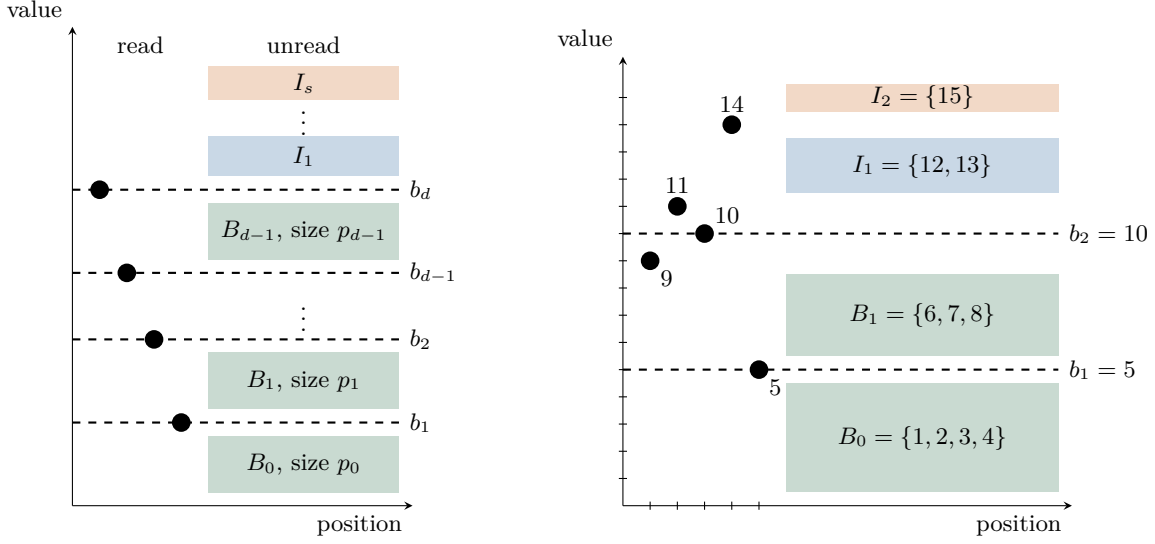


FIGURE 3. The bands and the stack. Left: the thresholds $b_1 < \dots < b_d$, the bands $B_0, \dots, B_{d-1}$ of unread base values with sizes $\mathbf{p} = (p_0, \dots, p_{d-1})$, and the stack $I_1, \dots, I_s$ above b_d . Right: the running example (4) for $d = 2$ after the prefix 9, 11, 10, 14, 5, with $b_1 = 5$, $b_2 = 10$, $\mathbf{p} = (4, 3)$, and the stack $\{12, 13\} \mid \{15\}$, as in row 5 of the table in example 4.2.

For example, in the permutation 316829574 the entries 3, 6 form an increasing 2-subsequence, and the subsequent projection onto values larger than 6 is 8, 9, 7, an occurrence of 231. Therefore 3, 6, 8, 9, 7 is an occurrence of $\beta_2 = 12453$. The interspersed values 2, 5, 4, all smaller than 6, do not belong to this projection.

4. THE LITERAL FAMILY RECURRENCE

In this section we write down the literal recurrence for β_d , show that it counts every avoider once, characterize the separations of its stack, and show that its state space is exponential. We now express the band updates in terms of their sizes. For $0 \leq i < d - 1$ and $0 \leq h < p_i$, define

$$T_{i,h}(\mathbf{p}) = (p_0, \dots, p_{i-1}, h, p_{i+1} + p_i - 1 - h, p_{i+2}, \dots, p_{d-1}). \quad (19)$$

The chosen letter becomes the new threshold b_{i+1} . Then h values remain in band i , and the values above it in the old band join band $i + 1$. For the last band put

$$U_h(\mathbf{p}) = (p_0, \dots, p_{d-2}, h), \quad \delta_h = p_{d-1} - 1 - h. \quad (20)$$

For $d = 1$, interpret (20) as $U_h((p_0)) = (h)$. Here the chosen letter is a new least d -trigger. The δ_h unread values above it and below the old b_d join the active head.

For example, take $d = 2$ and the prefix 9, 11, 10, 14, 5 of the running example (4), shown on the right of figure 3, with $b_1 = 5$, $b_2 = 10$, and $\mathbf{p} = (4, 3)$. Reading 2, which has $h = 1$ unread value of $B_0 = \{1, 2, 3, 4\}$ below it, makes 2 the new b_1 , leaves $\{1\}$ in band 0, and moves 3, 4 into band 1, so the control becomes $T_{0,1}(\mathbf{p}) = (1, 5)$. Reading 7 instead, which has $h = 1$ unread value of $B_1 = \{6, 7, 8\}$ below it, makes 7 the new b_2 and gives the control $U_1(\mathbf{p}) = (4, 1)$, and the $\delta_1 = 1$ value 8 joins the head, so the stack becomes $\{8, 12, 13\} \mid \{15\}$. Example 4.2 follows the whole running example through such moves.

Let $L = (\ell_1, \dots, \ell_s)$ be the sizes of the pending intervals, active head first. For a legal prefix (definition 2.4) with state $(\mathbf{p}, L)$, let $H_{\mathbf{p}}(L)$ be the number of completions that produce a β_d -avoiding permutation. By proposition 4.1(c) below, this number depends only on $(\mathbf{p}, L)$. For an empty stack we interpret $\ell_1 = 0$ and its tail as empty. Let $\text{nz}(x_1, \dots, x_r)$ denote the list obtained by deleting every zero entry.

Proposition 4.1 (State invariant). *Let σ be a legal prefix with thresholds $b_1 < \dots < b_d$, band sizes $\mathbf{p}$, and stack $\mathcal{S}(\sigma) = (I_1, \dots, I_s)$ with sizes $L = (\ell_1, \dots, \ell_s)$.*

- (a) *The stack consists of the unread values above b_d , so the ordered layout (18) holds, and a completion w of σ satisfies the residual obligations of all d -triggers of σ exactly when*

$$w|_{I_1 \cup \dots \cup I_s} \in \text{Av}(231)(I_1) \oplus \dots \oplus \text{Av}(231)(I_s). \quad (21)$$

- (b) *The legal moves from σ , and their effect on $(\mathbf{p}, L)$, are the following. A value of B_i with $i < d-1$ having h unread values of B_i below it is a non-trigger move to $(T_{i,h}(\mathbf{p}), L)$. A value of B_{d-1} having h unread values of B_{d-1} below it is a merger to $(U_h(\mathbf{p}), \text{nz}(\ell_1 + \delta_h, \ell_2, \dots))$. A value of I_1 of local rank r is a split to $(\mathbf{p}, \text{nz}(r-1, \ell_1 - r, \ell_2, \dots))$. No other letter is legal.*
- (c) *The set of β_d -avoiding completions of σ is determined, up to standardization, by $(\mathbf{p}, L)$.*

Proof. (a) By lemma 3.1, b_d is the least d -trigger of σ when it is nonvirtual, and σ has no d -trigger when b_d is virtual. Therefore, by definition 2.4, the stack consists of the unread values above b_d (no value exceeds a virtual $b_d = n + d$, so the two descriptions agree), while the unread values below b_d form the bands by definition. This is (18). Then (21) is proposition 2.10(a).

(b) We classify the unread letters by definition 2.4. Let $x \in B_i$ with $i < d-1$, and let h be the number of unread values of B_i below x . As observed before lemma 3.1, x ends an increasing subsequence of length $i+1$ but none of length $i+2 \leq d$, so x is not a d -trigger of σx : reading it is a non-trigger move, and the stack is unchanged. Replacing b_{i+1} by x leaves h values in band i and moves the other $p_i - 1 - h$ values into band $i+1$, giving $T_{i,h}(\mathbf{p})$. Next let $x \in B_{d-1}$, with h unread values of B_{d-1} below it. As observed before lemma 3.1 with $i = d-1$, the letter x ends an increasing d -subsequence (for $d = 1$, x is itself one) and changes only b_d . Since $x < b_d$, it is smaller than every d -trigger of σ by lemma 3.1. Reading it is therefore a merger, with $q = b_d$ when b_d is nonvirtual. When b_d is virtual, σ has no d -trigger and, by definition 2.4(a), E is the set of all unread values above x , again the unread values strictly between x and $b_d = n + d$. In both cases E has $\delta_h = p_{d-1} - 1 - h$ elements, x becomes the new b_d with h values left in the last band, and the stack becomes $\text{nz}(\ell_1 + \delta_h, \ell_2, \dots)$ by definition 2.4(a). Finally let $x \in I_1$ have local rank r . As noted after definition 2.4, x is a d -trigger of σx exceeding b_d , so reading it is a split. No threshold changes, since $x > b_d$, and the stack becomes $\text{nz}(r-1, \ell_1 - r, \ell_2, \dots)$ by definition 2.4(b). Every other unread letter lies in a deferred interval and is illegal by lemma 2.5(b).

(c) By proposition 2.10(c), the β_d -avoiding completions of σ are exactly the sequences of legal moves that read every unread letter, and by (b) the legal moves from a legal prefix, their successor states and their multiplicities depend only on $(\mathbf{p}, L)$ and on the local ranks of the letters read. Therefore, by induction on the number of unread letters, which drops by one at every move, the standardized set of avoiding completions depends only on $(\mathbf{p}, L)$. $\square$

For $(\mathbf{p}, L) \neq (\mathbf{0}, \emptyset)$, partitioning by the next letter gives the following expression for the number of completions that produce a β_d -avoiding permutation.

$$\begin{aligned}
H_{\mathbf{p}}(L) = & \sum_{i=0}^{d-2} \sum_{h=0}^{p_i-1} H_{T_{i,h}(\mathbf{p})}(L) \\
& + \sum_{h=0}^{p_{d-1}-1} H_{U_h(\mathbf{p})}(\text{nz}(\ell_1 + \delta_h, \ell_2, \dots)) \\
& + \min(2, \ell_1) H_{\mathbf{p}}(\text{nz}(\ell_1 - 1, \ell_2, \dots)) \\
& + \sum_{j=2}^{\ell_1-1} H_{\mathbf{p}}((j-1, \ell_1-j, \ell_2, \dots)).
\end{aligned} \tag{22}$$

The four lines represent, respectively, an early-band move, a last-band move, an endpoint choice in the active head, and an interior choice. The first double sum is empty when $d = 1$, and the last two lines are absent when the stack is empty. The initial and terminal conditions are

$$a_n^{(d)} := |\text{Av}_n(\beta_d)| = H_{(n,0,\dots,0)}(\emptyset), \quad H_{\mathbf{0}}(\emptyset) = 1. \tag{23}$$

For $d = 1$, identify $H_{(p)}(L) = W_p(L)$. The early-band sum is empty, $U_h((p)) = (h)$, and (22)–(23) reduce to (5)–(8).

Example 4.2 (What the second threshold adds). We specialize to $d = 2$ and scan the running example (4) again, which can be read down the second column of the table below. It avoids 12453, since 12453 contains 1342. The table lists, after each letter π_i , the type of move, the thresholds b_1, b_2 , the bands B_0, B_1 , the interval stack, and $H_{\mathbf{p}}(L)$, the number of 12453-avoiding completions. A dot marks a virtual threshold. Band 0 is the early band and band 1 the last band, and in the sense of definition 2.4 an early-band move is a non-trigger move and a last-band move is a merger. As in example 2.12, the table writes “merger” when the move joins values to the head or creates the stack, so that “last band” alone marks the case $E = \emptyset$, and it writes “head exhausted” for an endpoint choice in a head of size one.

i	π_i	move	b_1, b_2	B_0	B_1	interval stack	$H_{\mathbf{p}}(L)$
1	9	early band	9, ·	$\{1, \dots, 8\}$	$\{10, \dots, 15\}$	$\emptyset$	$H_{(8,6)}(\emptyset)$
2	11	last band, merger	9, 11	$\{1, \dots, 8\}$	$\{10\}$	$\{12, \dots, 15\}$	$H_{(8,1)}((4))$
3	10	last band	9, 10	$\{1, \dots, 8\}$	$\emptyset$	$\{12, \dots, 15\}$	$H_{(8,0)}((4))$
4	14	interior choice	9, 10	$\{1, \dots, 8\}$	$\emptyset$	$\{12, 13\} \mid \{15\}$	$H_{(8,0)}((2, 1))$
5	5	early band	5, 10	$\{1, \dots, 4\}$	$\{6, 7, 8\}$	$\{12, 13\} \mid \{15\}$	$H_{(4,3)}((2, 1))$
6	12	endpoint choice	5, 10	$\{1, \dots, 4\}$	$\{6, 7, 8\}$	$\{13\} \mid \{15\}$	$H_{(4,3)}((1, 1))$
7	6	last band, merger	5, 6	$\{1, \dots, 4\}$	$\emptyset$	$\{7, 8, 13\} \mid \{15\}$	$H_{(4,0)}((3, 1))$
8	2	early band	2, 6	$\{1\}$	$\{3, 4\}$	$\{7, 8, 13\} \mid \{15\}$	$H_{(1,2)}((3, 1))$
9	3	last band, merger	2, 3	$\{1\}$	$\emptyset$	$\{4, 7, 8, 13\} \mid \{15\}$	$H_{(1,0)}((4, 1))$
10	8	interior choice	2, 3	$\{1\}$	$\emptyset$	$\{4, 7\} \mid \{13\} \mid \{15\}$	$H_{(1,0)}((2, 1, 1))$
11	4	endpoint choice	2, 3	$\{1\}$	$\emptyset$	$\{7\} \mid \{13\} \mid \{15\}$	$H_{(1,0)}((1, 1, 1))$
12	7	head exhausted	2, 3	$\{1\}$	$\emptyset$	$\{13\} \mid \{15\}$	$H_{(1,0)}((1, 1))$
13	1	early band	1, 3	$\emptyset$	$\emptyset$	$\{13\} \mid \{15\}$	$H_{(0,0)}((1, 1))$
14	13	head exhausted	1, 3	$\emptyset$	$\emptyset$	$\{15\}$	$H_{(0,0)}((1))$
15	15	head exhausted	1, 3	$\emptyset$	$\emptyset$	$\emptyset$	$H_{(0,0)}(\emptyset)$

Compare with the one-threshold scan of example 2.12. The first letter 9 is not a 2-trigger and creates no obligation. It becomes b_1 , and the unread values above it form band 1. The first 2-trigger is 11, whose obligation constrains $\{12, \dots, 15\}$. The value 10 remains a base value, and reading it next makes it the least 2-trigger without joining anything to the head. The new minimum 5, a merger for $d = 1$, is here an early-band move: it transfers $\{6, 7, 8\}$ from band 0 to band 1 and leaves the head $\{12, 13\}$ unchanged. It is the 2-trigger 6, not the base value 5, that

joins $\{7, 8\}$ to the head. The read letters 9 to 12 lie inside the range of the head $\{7, 8, 13\}$, and by [corollary 4.4](#) below none of them separates 8 from 13: neither 9 nor 11 was read after an earlier 2-trigger smaller than 8, and 10 and 12 were read only after 2-triggers exceeding 8. The letter 14 separates 13 from 15, because it was read after the 2-trigger $10 < 13$. The letter 8, an endpoint choice for $d = 1$, is here an interior choice.

After the eighth letter, the thresholds are $b_1 = 2$ and $b_2 = 6$, the bands are $B_0 = \{1\}$ and $B_1 = \{3, 4\}$, and the stack is $\{7, 8, 13\} \mid \{15\}$. The state is $(\mathbf{p}, L) = ((1, 2), (3, 1))$. Its next choices are:

next	action	successor
1	update the first threshold	$H_{(0,2)}((3, 1))$
3	new 2-trigger, $\{4\}$ joins the head	$H_{(1,0)}((4, 1))$
4	new 2-trigger, no value is added to the head	$H_{(1,1)}((3, 1))$
7 or 13	endpoint choice in the active head	$H_{(1,2)}((2, 1))$, twice
8	interior choice with two singleton subintervals	$H_{(1,2)}((1, 1, 1))$
15	illegal until the head is exhausted	illegal

Therefore

$$H_{(1,2)}((3, 1)) = H_{(0,2)}((3, 1)) + H_{(1,0)}((4, 1)) + H_{(1,1)}((3, 1)) + 2H_{(1,2)}((2, 1)) + H_{(1,2)}((1, 1, 1)).$$

The extra threshold is visible in the first row: the choice 1 updates b_1 without creating a 2-trigger and, in this state, transfers no unread value to the second band. The values of an active head need not be read consecutively. Base values may occur between them. The scan continues with 3.

Theorem 4.3 (Literal recurrence). *For every $d \geq 1$, equations (22) and (23) count every β_d -avoiding permutation exactly once, as a maximal sequence of transitions from $((n, 0, \dots, 0), \emptyset)$.*

Proof. By [proposition 2.10\(c\)](#), the β_d -avoiding permutations are exactly the words all of whose prefixes are legal, and by [proposition 4.1\(b\)](#) the four types of terms in (22) are exactly the legal moves. Local ranks identify letters uniquely, except that two distinct endpoints have the same standardized successor and are recorded with multiplicity two. As in [section 2.3](#), the coefficient 2 stands for two transitions, labelled by the local ranks 1 and ℓ_1 . The quantity

$$\rho(\mathbf{p}, L) = \|\mathbf{p}\|_1 + \sum_{j=1}^s \ell_j \tag{24}$$

drops by one on every transition. An early-band move reduces the control mass by one. A last-band move reduces it by $\delta_h + 1$ and adds δ_h to the head. An active-head move consumes one head value. Therefore every maximal sequence of transitions ends at the unique zero state. Reading the chosen local ranks along such a path reconstructs one permutation, inverse to scanning that permutation. $\square$

The stack can be read off the scanned prefix. Recall that two unread values are adjacent if no unread value lies strictly between them. By [proposition 2.10\(c\)](#), the legal prefixes are exactly the prefixes of β_d -avoiding permutations, and every legal prefix has an avoiding completion, so condition (iii) below is not vacuous.

Corollary 4.4 (Separations of the stack). *After a legal prefix has been read, let $u < v$ be adjacent unread values above b_d . The following are equivalent.*

- (i) u and v lie in different intervals of the stack.
- (ii) Some letter x with $u < x < v$ was read after a d -trigger smaller than u .

(iii) u precedes v in every β_d -avoiding completion.

In particular, the stack is determined by the scanned prefix: it consists of the unread values above b_d , cut exactly at the adjacent pairs that satisfy (ii).

Proof. We prove that (ii) implies (iii), that (iii) implies (i), and that (i) implies (ii).

Suppose (ii) holds, and let c be the trigger. If v preceded u in a completion, an increasing d -subsequence ending at c , followed by x, v, u , would form β_d , since $c < u < x < v$. Therefore (iii) holds.

Suppose (i) fails, so that u and v lie in the same interval. The avoiding completion constructed in the proof of [proposition 2.10\(c\)](#) reads each interval in decreasing order, and so reads v before u . Therefore (iii) fails.

To see that (i) implies (ii), we show along the scan that (i) and (ii) agree for every adjacent pair above b_d . The induction repeats that of [lemma 2.11](#) with m replaced by b_d . We give it in full because of the additional early-band case. Initially the stack is empty and no trigger has been read. Let x be the letter read next. In an early-band move, which does not occur when $d = 1$, $x < b_d$ is not a d -trigger and lies below every unread value above b_d , so the stack and both conditions are unchanged. In a last-band move, x becomes the least trigger, and E joins the head or becomes the whole stack when the old stack was empty. An adjacent pair $u < v$ above x that was not an adjacent pair above the old b_d has $u \in E$, so u is smaller than every trigger read before x , and every letter between u and v was read before x . Therefore (ii) fails, and (i) fails because u and v lie in the single interval $E \cup I_1$, or E . All other pairs keep their members, their intervals, and the letters between them, and x lies below them, so both conditions are unchanged for them. In an active-head move the stack is nonempty, so b_d is a d -trigger by [lemma 3.1](#), and $x \in I_1$ is read after it. Only the unread neighbours $u < x < v$ of x , when both exist and $u > b_d$, can form a new adjacent pair. Then u lies in the part of I_1 below x , while v lies in the part above x or, if $x = \max I_1$, is the least element of I_2 . Thus u and v lie in different intervals, and x witnesses (ii) with the trigger $b_d < u$. All other pairs are unaffected. $\square$

Condition (ii) locates the separations, not the letters that produced them: several letters between u and v may satisfy (ii), and the letter whose split created the separation need not be among them. By [proposition 4.1\(c\)](#), the control $\mathbf{p}$ together with these separations determines the completions up to standardization. Adjacency is needed only in (ii), as the example after [lemma 2.11](#) shows. For arbitrary unread $u < v$ above b_d , conditions (i) and (iii) remain equivalent, and they hold exactly when some pair $a < c$ of adjacent unread values with $u \leq a < c \leq v$ satisfies (ii).

Proposition 4.5. *For fixed $d \geq 1$ and $n \geq d + 1$, literal memoization of (22) has at least F_{n-d} distinct composition arguments when computing $a_n^{(d)}$, where F_{n-d} is the $(n-d)$ th Fibonacci number, $F_1 = F_2 = 1$.*

Proof. Reading $1, 2, \dots, d$ first reaches the state $(\mathbf{0}, (n-d))$: the first $d-1$ letters are early-band moves, and d is a last-band move whose $n-d$ larger unread values form the head. The interior moves are then exactly those used in the proof of [proposition 2.16](#), with n there replaced by $n-d+1$. They reach every positive composition L satisfying

$$|L| + \text{len}(L) = n - d + 1,$$

and their number is

$$\sum_{s=1}^{\lfloor (n-d+1)/2 \rfloor} \binom{n-d-s}{s-1} = F_{n-d}. \quad \square$$

The literal state space is therefore exponential in n . [Section 5](#) removes the stack from the memoization key.

5. PROTECTED TAILS AND TRANSFER KERNELS

In this section we factor the literal recurrence through transfer kernels and derive recurrences for the kernels and for the empty-stack values. We apply [theorem 2.17](#) to the recurrence (22), with $\mathcal{C} = \mathbb{N}^d$ the set of controls and $\mathcal{B}$ the set of positive integers. We place the marker before an arbitrary tail L of the stack. An early-band move changes only the control. A last-band move can enlarge the active head, an endpoint can shorten it, and an interior letter can replace it by two parts. None of these transitions changes L before exposure, and (22) shows that the transitions leaving a state and their targets depend only on $\mathbf{p}$ and the blocks preceding the marker. Replacing L therefore gives the bijection of outgoing transitions required by (P2). The quantity (24) drops at every transition, so the recurrence terminates, and the only terminal state has an empty stack. Therefore no complete path can terminate before exposing L . Thus the hypotheses of [theorem 2.17](#) hold. We let $K_\ell^L(\mathbf{p}, \mathbf{t})$ be the number of stopped paths from $(\mathbf{p}, (\ell) \mid L)$ to $(\mathbf{t}, \emptyset \mid L)$.

Corollary 5.1 (Permutation-stack factorization). *The number $K_\ell^L(\mathbf{p}, \mathbf{t})$ is independent of L . We write $K_\ell(\mathbf{p}, \mathbf{t})$ for it. Then*

$$H_{\mathbf{p}}((\ell)L) = \sum_{\mathbf{t} \in \mathbb{N}^d} K_\ell(\mathbf{p}, \mathbf{t}) H_{\mathbf{t}}(L). \quad (25)$$

In particular, for $L = (\ell_1, \dots, \ell_s)$,

$$H_{\mathbf{p}}(L) = \sum_{\mathbf{t} \in \mathbb{N}^d} (K_{\ell_1} K_{\ell_2} \cdots K_{\ell_s})(\mathbf{p}, \mathbf{t}) H_{\mathbf{t}}(\emptyset). \quad (26)$$

As in [section 2.6](#), we call $K_\ell(\mathbf{p}, \cdot)$ a *kernel row*, with *source control* $\mathbf{p}$ and *terminal controls* $\mathbf{t}$, and its support is the set of $\mathbf{t}$ with $K_\ell(\mathbf{p}, \mathbf{t}) \neq 0$. The factorization replaces the composition argument by a product of transfer kernels, evaluated on the required source rows. The recurrence for the kernel rows follows, and [section 6](#) bounds the number of source and terminal controls required for coefficients through degree N .

Proposition 5.2 (Compressed kernel recurrence). *For $\ell \geq 1$, the transfer kernels satisfy*

$$\begin{aligned} K_\ell(\mathbf{p}, \mathbf{t}) = & \sum_{i=0}^{d-2} \sum_{h=0}^{p_i-1} K_\ell(T_{i,h}(\mathbf{p}), \mathbf{t}) \\ & + \sum_{h=0}^{p_{d-1}-1} K_{\ell+\delta_h}(U_h(\mathbf{p}), \mathbf{t}) + D_\ell(\mathbf{p}, \mathbf{t}) \\ & + \sum_{\substack{a,b \geq 1 \\ a+b=\ell-1}} \sum_{\mathbf{u} \in \mathbb{N}^d} K_a(\mathbf{p}, \mathbf{u}) K_b(\mathbf{u}, \mathbf{t}), \end{aligned} \quad (27)$$

where

$$D_1(\mathbf{p}, \mathbf{t}) = \mathbf{1}_{\mathbf{p}=\mathbf{t}}, \quad D_\ell(\mathbf{p}, \mathbf{t}) = 2K_{\ell-1}(\mathbf{p}, \mathbf{t}) \quad (\ell \geq 2). \quad (28)$$

Proof. We partition the stopped paths by their first move. Early- and last-band moves give the first two sums in (27). If $\ell = 1$, consuming the only head value exposes L at the same control. If $\ell \geq 2$, the two endpoint choices give the same smaller head. These are the terms in (28). An interior choice produces two parts of sizes $a, b \geq 1$ with $a + b = \ell - 1$. We move the marker to $(\mathbf{p}, (a) \mid (b)L)$. As in the scalar case, a path stopped at the first exposure of L passes through a unique first exposure of $(b)L$. Cutting there and summing over the intermediate control gives the double sum in (27). $\square$

We now connect the kernels to an empty stack. Put $K_0(\mathbf{p}, \mathbf{t}) = \mathbf{1}_{\mathbf{p}=\mathbf{t}}$ and $G_{\mathbf{p}} = H_{\mathbf{p}}(\emptyset)$.

Corollary 5.3 (Empty-stack recurrence). *The empty-stack values satisfy*

$$G_{\mathbf{p}} = \begin{cases} 1, & \mathbf{p} = \mathbf{0}, \\ \sum_{i=0}^{d-2} \sum_{h=0}^{p_i-1} G_{T_{i,h}(\mathbf{p})} + \sum_{h=0}^{p_{d-1}-1} \sum_{\mathbf{t} \in \mathbb{N}^d} K_{\delta_h}(U_h(\mathbf{p}), \mathbf{t}) G_{\mathbf{t}}, & \mathbf{p} \neq \mathbf{0}, \end{cases} \quad (29)$$

and

$$a_n^{(d)} = G_{(n,0,\dots,0)}. \quad (30)$$

Proof. This follows from (22) with an empty stack. A last-band move creates a head of size δ_h , to which (25) applies, and K_0 covers the case $\delta_h = 0$. The last equation is (23). $\square$

6. THE FAMILY ALGORITHM AND ITS COMPLEXITY

In this section we bound the supports of the kernel rows and count the operations, stored integers and bit sizes of the resulting algorithm, which proves [theorem 1.1](#). As in [section 2.6](#), we give a kernel source $(\mathbf{p}, \ell)$ the grade

$$w = \|\mathbf{p}\|_1 + \ell.$$

We verify below that increasing grade is a topological order for the kernel dependencies in (27), and that increasing control mass orders the empty-stack recurrence (29). The split term is controlled by the following support bound.

Lemma 6.1 (Kernel support). *For every $\ell \geq 1$,*

$$\text{supp } K_{\ell}(\mathbf{p}, \cdot) \subseteq \{\mathbf{p}\} \cup \{\mathbf{t} : \|\mathbf{t}\|_1 \leq \|\mathbf{p}\|_1 - 1\}. \quad (31)$$

Proof. Every base move consumes a letter and hence strictly lowers control mass. Active-head moves leave the control unchanged. If no base move occurs, the terminal control is $\mathbf{p}$. If at least one base move occurs, the terminal control has mass at most $\|\mathbf{p}\|_1 - 1$. $\square$

In the algorithm below, each row operation is coefficientwise, and every row not yet computed is zero.

Kernel phase. For $w = 1, 2, \dots, N$, and every $\mathbf{p} \in \mathbb{N}^d$, $\ell \geq 1$ with $\|\mathbf{p}\|_1 + \ell = w$, evaluate (27) coefficientwise and store the resulting sparse row. In each split term, for every $a, b \geq 1$ with $a + b = \ell - 1$ and $\mathbf{u} \in \text{supp } K_a(\mathbf{p}, \cdot)$, add the scalar multiple $K_a(\mathbf{p}, \mathbf{u}) K_b(\mathbf{u}, \cdot)$ to the row being computed.

Empty-stack phase. Set $G_0 = 1$. For $m = 1, 2, \dots, N$, and every $\mathbf{p} \in \mathbb{N}^d$ with $\|\mathbf{p}\|_1 = m$, evaluate (29), using K_0 when $\delta_h = 0$. Return $(G_{(n,0,\dots,0)})_{0 \leq n \leq N}$.

In the kernel phase, the early- and last-band rows in (27) have grade $w - 1$, for a last-band term because

$$\|U_h(\mathbf{p})\|_1 + (\ell + \delta_h) = \|\mathbf{p}\|_1 - (\delta_h + 1) + \ell + \delta_h = w - 1.$$

The term D_1 is initial data, while D_{ℓ} for $\ell \geq 2$ uses a row of grade $w - 1$. In a split term, $\|\mathbf{p}\|_1 + a < w$ and, by [lemma 6.1](#), $\|\mathbf{u}\|_1 + b \leq \|\mathbf{p}\|_1 + b < w$. Therefore increasing w is a topological order. In the empty-stack phase every terminal control used at mass m has smaller mass, including $U_h(\mathbf{p})$ when K_0 occurs, so increasing m is a topological order as well.

For the coefficients with $n \leq N$, we use the padded region

$$\mathbf{p} \in \mathbb{N}^d, \quad \ell \geq 1, \quad \|\mathbf{p}\|_1 + \ell \leq N. \quad (32)$$

The region contains exactly

$$\binom{N+d}{d+1} = O_d(N^{d+1}) \quad (33)$$

rows, by stars and bars. By [lemma 6.1](#), a row has at most $1 + \binom{\|\mathbf{p}\|_1 + d - 1}{d} = O_d(N^d)$ possible terminal controls.

Proof of [theorem 1.1](#). Correctness follows from [theorem 4.3](#) and [corollary 5.1](#): equations (27) and (29) are the first-step evaluations of the factorization (25), and the preceding topological orders evaluate all of their dependencies. We now bound the resources.

There are $O_d(N^{d+1})$ source rows and $O_d(N^d)$ possible entries per row, giving $O_d(N^{2d+1})$ stored integers. In a split term there are $O(N)$ choices of a , $O_d(N^d)$ intermediate controls $\mathbf{u}$, and $O_d(N^d)$ entries in the row added for each $\mathbf{u}$. Thus a sparse implementation uses $O_d(N^{2d+1})$ operations per source row and $O_d(N^{3d+2})$ over all kernels. The nonsplit part uses $O_d(N)$ row additions per source and therefore $O_d(N^{2d+2})$ operations in total. For each of the $O_d(N^d)$ empty-stack controls, (29) sums at most $O(N)$ choices over $O_d(N^d)$ terminal controls, so the G table costs $O_d(N^{2d+1})$. For $d \geq 1$, the exponents $2d + 2$ and $2d + 1$ are both smaller than $3d + 2$.

We call the quantity (24) of a state of (22) its *workload*. On a stopped kernel path the workload of a state is its control mass plus the sizes of the blocks before the marker. Each initial workload is at most N and decreases by one at every transition. At a state of (22), the number of outgoing transitions is the number of legal next letters and is at most the workload. Therefore a path has at most N transitions and at most N labeled choices at each nonterminal state. Each stored $G_{\mathbf{p}}$ counts complete paths in (22), while each stored kernel entry counts stopped kernel paths. Every such integer is at most $(N + 1)^N$ and has $O(N \log N)$ bits. Every intermediate product and partial sum is a nonnegative contribution to a stored value and obeys the same bound. Replacing integer multiplication by its $M(N \log N)$ bit cost proves the stated bit time. The bit storage is $O_d(N^{2d+2} \log N)$. $\square$

7. THE TWO-THRESHOLD QUOTIENT FOR 12453

In this section we specialize to $d = 2$ and write $\mathbf{p} = (p, q)$. Two symmetries of the kernels are available in this case, and they serve different purposes. The first is a translation in the lower threshold coordinate: a kernel entry depends on the first coordinates of its source and terminal controls only through their difference, so one coordinate can be dropped from the table. This is the symmetry that improves the $d = 2$ specialization $O(N^8)$ and $O(N^5)$ of [theorem 1.1](#) to the $O(N^7)$ and $O(N^4)$ of [corollary 7.4](#). The second is a partial translation in the upper coordinate. It leaves those exponents unchanged and halves the leading constant in the storage count. We first determine the support of a kernel row: the bound of [lemma 6.1](#) is attained exactly as follows.

Lemma 7.1 (Exact two-threshold support). *For every $\ell \geq 1$,*

$$\text{supp } K_\ell((p, q), \cdot) = \{(p, q)\} \cup \{(u, v) : 0 \leq u \leq p, v \geq 0, u + v \leq p + q - 1\}. \quad (34)$$

Proof. A base move never increases the first coordinate and strictly lowers the coordinate sum, which proves the inclusion. The terminal control (p, q) is reached by using no base move and reading endpoints until the head is exhausted. To reach a control (u, v) in the second set with $u < p$, first make the early-band move with $h = u$, which gives (u, Q) with $Q = p + q - 1 - u$. The support inequality says $v \leq Q$. If $v = Q$, make no further base move. If $v < Q$, make the last-band move with $h = v$. If $u = p$, then $v \leq q - 1$, and the last-band move with $h = v$ can be made directly. Finally, exhaust each active head by endpoint choices, which exposes the marked suffix at (u, v) . $\square$

We now prove the first translation.

Lemma 7.2 (First-coordinate translation). *For $\ell \geq 1$ and $0 \leq c \leq p$,*

$$K_\ell((p, q), (c, s)) = K_\ell((p - c, q), (0, s)). \quad (35)$$

Proof. Consider a kernel path ending at (c, s) . Its first control coordinate never falls below c , so the c lowest unread values are never read. Delete them throughout the path and shift all remaining values down by c . This preserves the active head sizes, every transition multiplicity, and the second control coordinate, and gives a path from $(p - c, q)$ to $(0, s)$. The inverse operation adds c unread values below every value used by the path. These added values are never selected, because the terminal first coordinate is c . They shift by exactly c the parameter h of every band-0 letter, leave the parameter h of every last-band letter and the local ranks within the active head unchanged, and preserve every transition multiplicity. The two operations are inverse. $\square$

We write

$$R_{\ell,a}(q, s) := K_{\ell}((a, q), (0, s)). \quad (36)$$

We call $R_{\ell,a}(q, s)$ a *reduced entry* and the table of all such entries the *first-coordinate quotient*. Throughout this section the letter a denotes a reduced first control coordinate, not a block size as in (5) and (27). Block sizes in a split are written ℓ_1, ℓ_2 . By lemma 7.2, every kernel entry is represented by one such reduced entry, with $a = p - c$. By lemma 7.1, for a reduced source (ℓ, a, q) with $a = 0$ the possible terminal coordinates are $0 \leq s \leq q$, and if $a > 0$ a base move is necessary and $0 \leq s \leq a + q - 1$. The first-coordinate quotient therefore has

$$\binom{N+2}{3} \text{ rows, } \frac{N(N+1)(N^2+N+4)}{12} \text{ entries.} \quad (37)$$

The entry formula follows by summing $q + 1$ for $a = 0$ and $a + q$ for $a > 0$ over the same padded source region. In particular, it already gives $O(N^4)$ storage. We count the operations after the second translation.

We now rewrite the kernel recurrence in the reduced variables by applying lemma 7.2 to (27). An early-band move from (a, q) gives $R_{\ell,h}(a + q - h - 1, s)$, and a last-band move with parameter r , written r here because h indexes the early-band sum, gives $R_{\ell+q-r-1,a}(r, s)$. In a split with intermediate control (c, m) , the first factor $K_{\ell_1}((a, q), (c, m))$ vanishes unless $c \leq a$, since both parts of the support (34) have first coordinate at most a . For $0 \leq c \leq a$ the two factors become

$$R_{\ell_1, a-c}(q, m) R_{\ell_2, c}(m, s).$$

Writing $a_1 = a - c$ and $a_2 = c$ therefore gives the reduced recurrence

$$\begin{aligned} R_{\ell,a}(q, s) = & \sum_{h=0}^{a-1} R_{\ell,h}(a + q - h - 1, s) + \sum_{r=0}^{q-1} R_{\ell+q-r-1,a}(r, s) + D_{\ell,a}(q, s) \\ & + \sum_{\substack{\ell_1, \ell_2 \geq 1 \\ \ell_1 + \ell_2 = \ell - 1}} \sum_{\substack{a_1, a_2 \geq 0 \\ a_1 + a_2 = a}} \sum_{m \geq 0} R_{\ell_1, a_1}(q, m) R_{\ell_2, a_2}(m, s), \end{aligned} \quad (38)$$

where

$$D_{1,a}(q, s) = \mathbf{1}_{a=0, q=s}, \quad D_{\ell,a}(q, s) = 2R_{\ell-1,a}(q, s) \quad (\ell \geq 2),$$

and entries outside the support above are zero.

We call $\ell + a + q$ the *reduced grade* of $R_{\ell,a}(q, \cdot)$. Increasing reduced grade is a topological order for (38). Every term of the first two sums and every D term has reduced grade $\ell + a + q - 1$. In a split, the first factor has reduced grade $\ell_1 + a_1 + q \leq \ell + a + q - 2$. A nonzero first factor forces $m \leq q$ when $a_1 = 0$ and $m \leq a_1 + q - 1$ when $a_1 > 0$, so the second factor has reduced grade $\ell_2 + a_2 + m \leq \ell + a + q - 2$.

The second control coordinate admits a partial quotient as well: raising both the second source coordinate and the terminal coordinate by one leaves a reduced entry unchanged whenever the terminal coordinate is at least $a - 1$.

Lemma 7.3 (Second-coordinate translation). *For $\ell \geq 1$ and $a, q, s \geq 0$, whenever $s \geq a - 1$,*

$$R_{\ell,a}(q+1, s+1) = R_{\ell,a}(q, s). \quad (39)$$

Proof. Let $\gamma = \ell + a + q + 1$ be the reduced grade of the left-hand side. We argue by induction on $\gamma \geq 2$. If $\gamma = 2$, then $(\ell, a, q) = (1, 0, 0)$, and (38) gives

$$R_{1,0}(1, s+1) = \mathbf{1}_{s=0} = R_{1,0}(0, s).$$

Assume henceforth that $\gamma > 2$ and that the claim holds at every smaller reduced grade. The h th term of the first sum on the left is $R_{\ell,h}(a+q-h, s+1)$. Since $s \geq a-1 \geq h-1$, induction translates this to $R_{\ell,h}(a+q-h-1, s)$, the corresponding term on the right.

In the second sum, the $r = 0$ term on the left vanishes: at second source coordinate zero the largest possible terminal coordinate is 0 if $a = 0$, and $a - 1$ otherwise, while the terminal coordinate satisfies $s+1 \geq 1$ and $s+1 \geq a$. Put $r = j+1$ in the remaining terms. Induction gives

$$R_{\ell+q-j-1,a}(j+1, s+1) = R_{\ell+q-j-1,a}(j, s),$$

which gives exactly the terms $0 \leq j < q$ on the right. The D terms also match: for $\ell = 1$ both are $\mathbf{1}_{a=0, q=s}$, while for $\ell > 1$ the claim follows from induction applied to $2R_{\ell-1,a}$.

We now compare one split with $a = a_1 + a_2$. On the left it contributes

$$\sum_{m \geq 0} R_{\ell_1, a_1}(q+1, m) R_{\ell_2, a_2}(m, s+1).$$

The $m = 0$ term vanishes by the support of the second factor. Put $m = k+1$. If that second factor is nonzero, then $k \geq a_1 - 1$: when $a_2 = 0$, support gives $k \geq s \geq a_1 - 1$, while for $a_2 > 0$ it gives $k \geq s+1 - a_2 \geq a_1$. Also $s \geq a-1 \geq a_2 - 1$. Thus the induction hypothesis applies to both factors.

Both induction calls are at smaller reduced grade. The first has grade $\ell_1 + a_1 + q + 1 < \gamma$. Nonvanishing of that factor also gives

$$k \leq q \quad \text{if } a_1 = 0, \quad k \leq a_1 + q - 1 \quad \text{if } a_1 > 0,$$

by (34). Therefore the second factor has reduced grade $\ell_2 + a_2 + k + 1 < \gamma$. Induction then gives

$$R_{\ell_1, a_1}(q+1, k+1) R_{\ell_2, a_2}(k+1, s+1) = R_{\ell_1, a_1}(q, k) R_{\ell_2, a_2}(k, s).$$

The same upper and lower support bounds hold for every nonzero term on the right, so this is a bijective reindexing of the split sum. $\square$

The hypothesis $s \geq a - 1$ cannot be weakened: equation (38) gives $R_{1,2}(1, 1) = 3$ but $R_{1,2}(0, 0) = 2$.

For $a = 0$ and $0 \leq s \leq q$, lemma 7.3 gives $R_{\ell,0}(q, s) = R_{\ell,0}(q-s, 0)$. For $a \geq 1$ and $s \geq a$, it gives

$$R_{\ell,a}(q, s) = R_{\ell,a}(q-s+a-1, a-1),$$

where the support condition ensures that the new second source coordinate is nonnegative. Thus we retain only terminal coordinate $s = 0$ when $a = 0$, and terminal coordinates $0 \leq s < a$ when $a \geq 1$. The exact number of retained entries is

$$\binom{N+1}{2} + \binom{N+2}{4} = \frac{N(N+1)(N^2+N+10)}{24}. \quad (40)$$

Thus the second translation halves the leading storage constant relative to the first-coordinate quotient. Exact polynomials for loop iterations and stored entries are recorded in the computation report in the supplement.

Corollary 7.4 (Sharpened 12453 bound). *The numbers $|\text{Av}_n(12453)|$ for $0 \leq n \leq N$ can be computed using $O(N^7)$ integer arithmetic operations and $O(N^4)$ stored integers. The bit time is $O(N^7 M(N \log N))$, and the bit storage is $O(N^5 \log N)$, with M as in [theorem 1.1](#).*

Proof. We count the operations for the first-coordinate quotient, evaluated in increasing reduced grade, which the preceding paragraphs showed to be a topological order for (38). It has $O(N^3)$ source rows and $O(N^4)$ stored entries by (37). Evaluating the split terms involves $O(N)$ choices for the block split ℓ_1 , $O(N)$ choices for the control split a_1 , and $O(N)$ possible intermediate second coordinates m . For each such triple, adding the row $R_{\ell_2, a_2}(m, \cdot)$ uses $O(N)$ possible terminal coordinates. Thus one source row costs $O(N^4)$ operations and all split terms cost $O(N^7)$. The second-coordinate translation can reduce the storage constant. Each omitted entry is recovered from (39) in $O(1)$ operations. Evaluating the nonsplit kernel terms costs at most $O(N^5)$. The empty-stack table has $O(N^2)$ source controls, $O(N)$ last-band choices, and $O(N^2)$ possible terminal controls, so it also costs $O(N^5)$. The workload argument in the proof of [theorem 1.1](#) gives $O(N \log N)$ bits per integer, which gives the stated bit bounds. $\square$

8. THE COUNTING SEQUENCE OF $\text{Av}(12453)$

In this section we describe the computation of $|\text{Av}_n(12453)|$ for $n \leq 150$ and, in [section 8.1](#), uniform random generation. [Table 1](#) lists some of the values. The file `code/data/av12453_term s_0_150.txt` in the supplement contains all of them.

We evaluated the reduced recurrence modulo pairwise distinct primes and reconstructed the values over the integers. The programs `av12453_fast_rns.cpp` and `reconstruct_av12453_rns.py` in the supplement perform the modular evaluation, the check of the coefficient bound, and the reconstruction. Complete source, execution logs, residues, and machine-readable output are included in the supplement.

The reconstruction rests on a bound for the coefficients. Put $b_m = |\text{Av}_m(1342)|$ and

$$B_N := \sum_{m=0}^N \binom{N}{m}^2 b_m. \quad (41)$$

As Bóna [10] observed, deleting the left-to-right minima of a 12453-avoider leaves a 1342-avoider: a 1342 among the remaining entries could be preceded by an earlier smaller entry, producing a 12453. If m entries remain, choose their positions, their values, and their standardization. The deleted minima occupy the remaining positions, and their remaining values are forced in decreasing order. Therefore $|\text{Av}_N(12453)| \leq B_N$.

The integers b_m are computed from (17). The source `av12453_fast_rns.cpp` implements (38) and the empty-stack recurrence (29) modulo each selected prime. By [theorem 4.3](#), [corollaries 5.1](#) and [5.3](#), [proposition 5.2](#), and [lemmas 7.2](#) and [7.3](#), its outputs are the residues of the coefficients. The moduli are the twenty-four largest primes below 2^{31} , recorded with the residue packs in the supplement. The program `reconstruct_av12453_rns.py` verifies that the product $\mathcal{M}$ of the eighteen largest exceeds B_{150} . The margin is $\mathcal{M}/B_{150} > 1.92$. Since the nonnegative bound B_N is nondecreasing, every coefficient through degree 150 lies in $[0, \mathcal{M})$. The Chinese remainder theorem therefore determines each coefficient uniquely.

The six remaining primes, withheld from the reconstruction, agree with every reconstructed coefficient. A separately written single-modulus $d = 2$ program computes all 151 coefficients modulo $2^{61} - 1$ and also agrees. These are error-detection checks, not inputs to the uniqueness argument. The values through $n = 38$ agree with those previously known, in OEIS A116485 [20], which come from Biers-Ariel's program [7, 8].

n	$ \text{Av}_n(12453) $
0	1
1	1
2	2
3	6
4	24
5	119
6	694
7	4581
8	33286
9	260927
$\vdots$	$\vdots$
70	272047014426505848053184970016495041863317183438034114760726065251824322
71	3650982161839069245244704534509742637654076726349158226492884931736173219
72	49052479122475377086437757538549643339461005664517035047738861064660650462
73	659758130045267152279725187963519417028655461010134067851632959874299139314
74	8883183490302999538631811574792997430948698470130245020946152645591389765992
75	119729358528034563299140115867589732584037184044497744461031132334923227197902
76	1615360029787371109415151060029379769950586589470709389115337666950300899664892
77	21815422062572671915613939537695855501209448778205442702933011769377618433242686
78	294898723790650390962604916545121449539951614378915249501191039739041483852959800
79	3990128916983023825344047432227024955428537983460787346271686479611600994645296610
80	54037578151511512886697895933429730101506005764295760914601440348390527553113513544
$\vdots$	$\vdots$
140	132563129209678900097059991729972685485162293985609301526390639958435284778009278831298106 90309462018830746612767544034249954911304218723154384122701096
141	185288351260158636665154188074881299733908503673301584716105567137519163453156092505084472 656884441044425731642146730167064793041572656010317336745260230
142	259061502229303438137253209397747821095567080423357798067579823149354198936311384115189454 8670967420193329587946042347831801662279707120672955517288773276
143	362314147509666008769192778238021298134479862253752577900601633529337398739612132874209755 00425725343682999003885891306094188761363063724056324785823177873
144	506866570108080992172827702390824721485314592184236068629394550709633572784507558031741140 313981790054699490480890332852463168535779082217300140047632872622
145	709293982764754919593192205776244504591725854558410034064407257992516450166543645900069113 833616956508257221756849768300959452695409003062226217099351149132
146	992845185957788995426922964104645694631352710528923383764487517996963854569544958714672498 70870043899715835749168084058967962235009591429889624809811996930400
147	13901376928245085643377786818058778546882793194476967952737658048244365801756110716322666 2022575277081761365776749741699180828349495930184915117643473649518426
148	194694442948917259391570834796944420652677476522578329510621571835317403885469519572114751 35962040032085429461259232291433428857567215495002695620164810628780672
149	272751533859863976347598705423927955482798527767963594985771513751414865658702668306436986 953076260565473100844770203223196420613511632988251274688398628136581935
150	382205773917840409900771979163202054515626683096917522113147052334208885924065532828958769 7088297741511203960758500758477699391695509589920898592073159171767612574

TABLE 1. The numbers $|\text{Av}_n(12453)|$ for $0 \leq n \leq 9$, $70 \leq n \leq 80$, and $140 \leq n \leq 150$.

8.1. Uniform random generation. The transfer tables can also be used to generate uniformly random avoiders. By [theorem 4.3](#), the labeled complete paths of (22) from the initial state $H_{(n,0,\dots,0)}(\emptyset)$ correspond bijectively to the β_d -avoiding permutations of length n . By the recursive method [19], such a path is uniformly random when every transition is chosen with probability proportional to the number of complete paths through it, that is, to the value of the corresponding summand of (22). That value is a completion count $H_{\mathbf{p}}(L)$ for a stack L of arbitrary length, which the factorization (26) expresses through the stored kernels. We therefore sample a complete path of the factorized recurrence itself. The first-move partitions in the proofs of [proposition 5.2](#)

and [corollary 5.3](#) identify the summands of (27) and (29) with a partition of the stopped and complete paths. Iterating the cut-and-concatenate bijection of [theorem 2.17](#) maps these paths onto the complete paths of (22). At an empty-stack state, a summand of (29) is chosen with probability proportional to its value. This selects a base letter and, for a last-band move, the terminal control $\mathbf{t}$ at which the new head will have been exhausted. Inside a kernel $K_\ell(\mathbf{p}, \mathbf{t})$, a summand of (27) is chosen in the same way. This selects a base move, an endpoint of the active head, or an interior letter of it together with the intermediate control $\mathbf{u}$ of the split, and each pending block carries the control at which it must be exposed. By induction on the grade, a given stopped path of $K_\ell(\mathbf{p}, \mathbf{t})$ is produced with probability $1/K_\ell(\mathbf{p}, \mathbf{t})$ and a given complete path from $G_{\mathbf{p}}$ with probability $1/G_{\mathbf{p}}$. At $\mathbf{p} = (n, 0, \dots, 0)$ this is $1/|\text{Av}_n(\beta_d)|$.

Proposition 8.1 (Uniform random generation). *Let $N \geq n \geq 0$. Once the first-coordinate quotient $R_{\ell,a}(q, s)$ of [section 7](#) and the empty-stack values $G_{(p,q)}$ have been computed for size N , a uniformly random element of $\text{Av}_n(12453)$ can be produced with $O(n^4)$ arithmetic operations on the stored integers.*

Proof. The paragraph before the proposition shows that the distribution is uniform. A permutation of length n is produced by n letter choices. Each choice scans the summands of one instance of (29) or (27) in a fixed order and stops at the first summand whose running total exceeds a uniformly random integer U with $0 \leq U < T$, where T is the stored value of the state. Every summand is one stored entry, twice one, or a product of two, by [lemma 7.2](#). Every state reached from $(n, 0)$ has grade at most n . At an empty-stack state the last-band terms range over the band position and the terminal control, $O(n^3)$ summands. Inside a kernel the interior terms range over the split position and the intermediate control, again $O(n^3)$. The other groups are smaller. Therefore each choice costs $O(n^3)$ operations and a sample costs $O(n^4)$. $\square$

Unlike the coefficients of [table 1](#), our sampler runs in floating point. [Proposition 8.1](#) is the exact-arithmetic statement. [Appendix A](#) describes the implementation and proves that, for $n \leq 300$, its output is within total variation distance $3.5 \cdot 10^{-5}$ of the uniform distribution on $\text{Av}_n(12453)$ ([proposition A.2](#)). Every sample is verified to avoid 12453 by an independent test based on [lemma 2.1](#), and the sampler was checked against exhaustive enumeration for small n . Building the tables for $N = 300$ took about four hours on eight cores. One million samples of length 300 then took about thirty-five minutes.

[Figure 4](#) shows the resulting position–value heatmap, rendered with the script that draws the heatmaps of the PermPAL database [6, 1], beside one of the samples. The first letter has mean 291.9 over the sample. The dark band from the top left to the bottom right shows that a typical avoider stays close to a decreasing curve. The left-to-right minima, whose deletion leaves a 1342-avoider, as Bóna [10] observed, lie below it, and their mean number is 77.5. The single sample shows the same features: a decreasing band of letters with the left-to-right minima scattered below it, and above it the letters exceeding the first letter, which form a 1342-avoider.

9. FORMAL VERIFICATION

The counting recurrences of this paper for $d \leq 2$ have been verified in the Lean 4 proof assistant [13] with the Mathlib library [24]. The development is in the directory `formal/` of the supplement. Its four main theorems state that, for every n ,

$$\begin{aligned} W_n(\emptyset) &= |\text{Av}_n(1342)|, & G_n &= |\text{Av}_n(1342)|, \\ H_{(n,0)}(\emptyset) &= |\text{Av}_n(12453)|, & G_{(n,0)} &= |\text{Av}_n(12453)|. \end{aligned}$$

That is, [theorem 4.3](#) holds for $d \leq 2$, and the recurrences evaluated by the algorithm of [section 6](#) count $\text{Av}_n(1342)$ and $\text{Av}_n(12453)$.

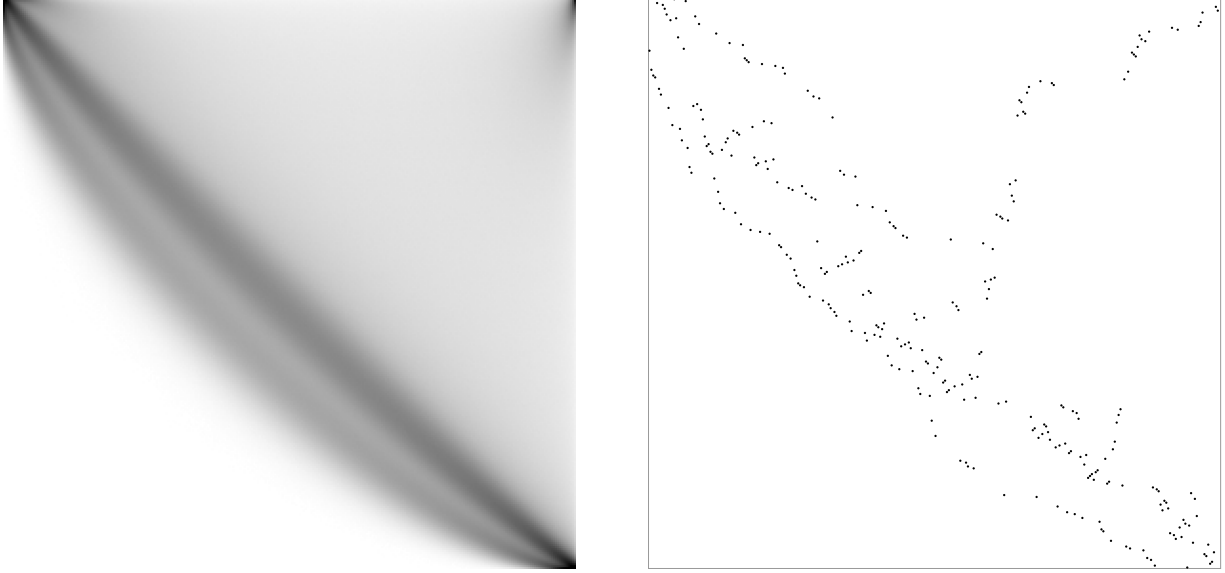


FIGURE 4. Left: position–value heatmap of one million elements of $\text{Av}_{300}(12453)$ drawn by the floating-point sampler. The cell at position i (left to right) and value v (bottom to top) is shaded by the square root of the number of samples with $\pi_i = v$, from white (no sample) to black (the largest cell count, 70 204). Right: one of the samples, with a dot at (i, π_i) in the same orientation. Its first letter is 272 and it has 71 left-to-right minima.

A reader who wants to verify this needs to do two things. The first is to run the script `check.sh`. It builds the development and checks that it contains no unproved statement and that every theorem depends only on Lean’s three standard axioms. The second is to compare with this paper the definitions that occur in the four statements: pattern avoidance, the patterns β_d , and the recurrences. These definitions occupy about seventy lines of Lean. The proofs need not be read.

The proofs follow the paper. The state of a scanned prefix is defined from the prefix, the separation criterion of [lemma 2.11](#) and [corollary 4.4](#) is the inductive invariant, and the recurrences follow by induction on the number of unread values. The factorization (25) is proved from the equations of H , without the general [theorem 2.17](#). Not formalized are the complexity statements of [theorem 1.1](#) and [corollary 7.4](#), the translation quotients of [section 7](#) and hence the programs of [section 8](#), [theorem 2.17](#) in its general form, and the family for $d \geq 3$. [Appendix B](#) describes the development in more detail.

10. CONSEQUENCES AND OUTLOOK

In this section we place 12453 among the length-five classes and state a conjecture on the asymptotics of its counting sequence. Two patterns are *Wilf-equivalent* if they have equally many avoiders of every length. Reverse, complement, and inverse preserve all counting sequences. The dihedral orbit of 12453 has eight elements, and the prefix-reversal equivalence of Backelin–West–Xin [5] gives $12453 = 12 \oplus 231 \sim 21 \oplus 231 = 21453$, whose orbit supplies eight more. Therefore the algorithm, bounds, and coefficients apply to all sixteen members of the length-five Wilf class represented by 12453.

Bóna determined the Stanley–Wilf limit of every member of the family [10]:

$$\lim_{n \rightarrow \infty} (a_n^{(d)})^{1/n} = (d - 1 + \sqrt{8})^2.$$

For the remainder of this section, put $a_n = a_n^{(2)} = |\text{Av}_n(12453)|$ and $\mu = (1 + \sqrt{8})^2 = 9 + 4\sqrt{2}$. The coefficients through $n = 100$, read with the series analysis of Clisby, Conway, Guttmann and Inoue [12], suggest the following refinement.

Conjecture 10.1. *There are constants $C, \kappa > 0$ and $h \in \mathbb{R}$ such that, as $n \rightarrow \infty$,*

$$a_n = C\mu^n e^{-\kappa n^{1/3}} n^{-17/4} \left(1 + hn^{-1/3} + O(n^{-2/3})\right). \quad (42)$$

The exponents $1/3$ and $-17/4$ were selected using only $a_0, \dots, a_{100}$. The parameters $\kappa, \log C, h$ were obtained by unweighted least squares on $70 \leq n \leq 100$ for the model $\log a_n = n \log \mu - \kappa n^{1/3} - \frac{17}{4} \log n + \log C + hn^{-1/3}$, and were fixed before $a_{101}, \dots, a_{150}$ were computed. The frozen values are $(\kappa, C, h) = (1.34550865, 0.82710184, 1.36800770)$ to the digits shown. On those fifty excluded coefficients, the maximum absolute difference between the observed and predicted values of $\log a_n$ is below $4.3 \cdot 10^{-7}$. The fit and this check are reproduced by a program in the supplement. From shorter series, the 2022 analysis estimated the stretched-exponential base $e^{-\kappa}$ (their μ_1) between 0.26 and 0.29 and the power-law exponent as -4.67 ± 0.1 [12]. The frozen fit gives $e^{-\kappa} \approx 0.260$, at the lower end of that range. Its exponent $-17/4$ differs from the 2022 estimate, but we do not regard the two as in conflict. On $70 \leq n \leq 150$ the exponent $-17/4$ fits the model with one correction term far better than -4.67 does. The preference shrinks by more than an order of magnitude once a second correction term $n^{-2/3}$ is admitted. In fits with two correction terms, a change in the stretched exponent $1/3$ is largely compensated by changes in the power-law exponent and in κ . The inferred power-law exponent is therefore sensitive to the correction terms included, and the available coefficients do not determine it reliably. We record $-17/4$ as the simplest value consistent with the exact coefficients, not as a sharpening of the 2022 estimate. This supports [conjecture 10.1](#) but proves neither it nor any D-finiteness claim.

DATA AND CODE AVAILABILITY

The code, coefficients, residues, reconstruction script, and computation reports used for [section 8](#), the sampler and heatmap tools of [section 8.1](#), the Permuta walkthrough with its exhaustive checks for length at most 8, and the Lean development of [section 9](#) and [appendix B](#) are in the public repository

<https://github.com/ulfarsson/public-12453>

which is the supplement referred to throughout the paper. The code, data, and Lean development are released under the Apache License 2.0. The repository's `MANIFEST.sha256` records the SHA-256 digest of every file, and its README files give verification, reconstruction, and audit commands.

STATEMENT ON THE USE OF AI

Large language model assistants (Claude Opus and Claude Fable from Anthropic, and GPT Sol from OpenAI) were used throughout the preparation of this paper under the direction of the author: to draft and revise the text, to write and test the companion programs and run the computations reported in [section 8](#), and to develop the Lean formalization of [section 9](#). The mathematical content was specified and checked by the author. The computational claims are supported by the independent implementations and checks described in [section 8](#) and, for the sampler, by the error bound of [appendix A](#), and the statements of [section 9](#) are verified by the Lean kernel. In addition, the Permuta walkthrough in the supplement checks every combinatorial statement of the paper that has finite content exhaustively for all permutations of length at most 8, by comparing the paper's constructions and displayed formulas with brute-force computations from the definitions. These checks are written with the Permuta library [21], whose code predates

the large language model assistants, and they do not use the implementations described in [section 8](#). The author takes full responsibility for the paper.

APPENDIX A. FLOATING-POINT SAMPLING

This appendix describes the floating-point implementation of the sampler of [proposition 8.1](#) and bounds the distance of its output distribution from uniformity. Throughout, the tables have size $N = 300$ and $n \leq N$.

A.1. The implementation. For $d = 2$ and a control $(p, q) \neq (0, 0)$, the empty-stack recurrence (29) reads, in the reduced entries,

$$G_{(p,q)} = \sum_{h=0}^{p-1} G_{(h,p+q-1-h)} + \mathbf{1}_{q \geq 1} G_{(p,q-1)} + \sum_{h=0}^{q-2} \sum_{c=0}^p \sum_{t \geq 0} R_{q-1-h,p-c}(h,t) G_{(c,t)}. \quad (43)$$

The middle term is the last-band move with $\delta_h = 0$. At $N = 300$ the largest values are about 2^{1115} , beyond the largest binary64 number, which is below 2^{1024} . The implementation therefore stores the scaled values

$$R'_{\ell,a}(q,s) = 2^{-2(\ell+a+q)} R_{\ell,a}(q,s), \quad G'_{(p,q)} = 2^{-2(p+q)} G_{(p,q)}.$$

In the scaled values, every band term and every D term of (38) is multiplied by $1/4$, the split term becomes $4^{m-1} R'_{\ell_1,a_1}(q,m) R'_{\ell_2,a_2}(m,s)$, and $D_{1,a}(q,s)$ becomes $2^{-2(1+a+q)} \mathbf{1}_{a=0,q=s}$. In (43) the first two terms are multiplied by $1/4$ and the last becomes $4^{t-1} R'_{q-1-h,p-c}(h,t) G'_{(c,t)}$. In the build for $N = 300$ the nonzero scaled values lie between 2^{-600} and 2^{515} , inside the normal range of binary64, and none is infinite or subnormal.

A term $4^{m-1} r_1 r_2$ with scaled factors r_1, r_2 is formed as $(2^{m-1} r_1)(2^{m-1} r_2)$, or as $(4^{m-1} r_1) r_2$ when $4^{m-1} r_1 \leq 10^{300}$. The factors are then at least 2^{-603} , whereas $r_1 r_2$ can be as small as 2^{-1200} and would underflow. The program computes the reduced entries in increasing reduced grade and then the values $G'_{(p,q)}$ in increasing $p+q$. It forms each value by adding its summands to a running sum, one at a time, in a fixed order, starting from zero. In (43) the terms with fixed h and c are first added into an inner sum over t . The compiler may fuse a multiplication with the addition that follows it.

The sampler follows [proposition 8.1](#). Its state is the control (p, q) together with the stack of pending blocks, each carrying its size and the control at which it must be exposed. At an empty-stack state the candidates of a letter choice are the summands of (43), and the stored value of the state is $G'_{(p,q)}$. At a state whose head has size ℓ and is to be exposed at control (c, t) , the stored value is $R'_{\ell,a}(q, t)$ with $a = p - c$, and the candidates are the summands of (38). For $\ell \geq 2$ the term $D_{\ell,a}(q, t)$ is two candidates, one for each endpoint of the head. Each candidate is given its value in the scaled recurrence, so that the candidates of a state sum to its stored value.

The candidates are scanned in a fixed order, in groups. Each band term and each endpoint is a group of its own. At an empty-stack state the terms of the last sum of (43) with fixed h and c form one group, and inside a head the split terms with fixed ℓ_1 and a_1 form one group. The sampler computes the sum of each group by adding its members in order, and the running sums $A_1 \leq A_2 \leq \dots$ of the group sums by adding these in order. It takes a multiple x of 2^{-53} in $[0, 1)$ from 53 random bits and rounds $x\hat{T}$ to y , where $\hat{T}$ is the stored value of the state. It then takes the first group g with $A_g > y$ and, inside that group, the first member whose running sum exceeds the rounded difference $y - A_{g-1}$, or the last member if there is none. Candidates of value zero are skipped. If no group has $A_g > y$, the sampler scans again with y multiplied by $A/\hat{T}$, where A is the total of the scan, and if that also falls short it takes the last candidate of positive value.

The floating-point values $G_{(n,0)}$ agree with the coefficients of [table 1](#) through $n = 150$ to about 10^{-11} relative error.

A.2. The error bound. We write $\text{fl}(z)$ for z rounded to binary64, $u = 2^{-53}$ for the unit roundoff and $\gamma_m = mu/(1 - mu)$ for $mu < 1$. If z is the sum or the product of two binary64 numbers, or a product plus a number in a fused operation, and $|z|$ lies in the normal range, then $\text{fl}(z) = z(1 + \delta)$ with $|\delta| \leq u$. A multiplication by a power of two is exact when the result is normal. If $|\theta_j| \leq \gamma_j$ and $|\theta_k| \leq \gamma_k$, then $(1 + \theta_j)(1 + \theta_k) = 1 + \theta$ with $|\theta| \leq \gamma_{j+k}$, and a product of m factors $1 + \delta_i$ with $|\delta_i| \leq u$ is $1 + \theta$ with $|\theta| \leq \gamma_m$ [16, Lemmas 3.1 and 3.3].

For $R_{\ell,a}(q, s)$ put $\lambda = \ell + a + q - s$, and for $G_{(p,q)}$ put $\lambda = p + q$. This is the number of letters read by the stopped paths of $R_{\ell,a}(q, s)$ and by the complete paths from $G_{(p,q)}$, and $\lambda \leq N$. Every band term and every D term has one letter fewer. The two factors of a split term together have $\lambda - 1$ letters, since the split reads one letter itself, and the same holds for the two factors of a term of the last sum of (43). Put $\nu = 3\,352\,801$. Counting the summands of (38) for fixed s , and of (43), shows that no value has more than $\nu - 1$ summands and no letter choice has more than $\nu - 1$ candidates. The maximum is attained by $G_{(149,151)}$.

Lemma A.1 (Accuracy of the tables). *Every stored value is its scaled exact value times $1 + \theta$ with $|\theta| \leq \gamma_{\nu\lambda}$. In particular, a stored value is zero if and only if its exact value is zero.*

Proof. We argue by induction on λ , in the order in which the values are computed. A nonzero exact scaled value is a positive integer times 2^{-2w} , where $w \leq N$ is the reduced grade or $p + q$, so it is at least 2^{-600} , and so is every nonzero summand. Since the summands are nonnegative and rounding is monotone, every running sum lies between its first nonzero summand and the stored value. With the range of the stored values and the form of the products above, every nonzero number formed lies between 2^{-603} and 2^{997} , so no underflow or overflow occurs.

A summand that is one stored value times a power of two is formed without rounding. A product of two stored values is formed with one rounding, or with none if it is fused with the next addition. By the induction hypothesis and since the letters of the factors add up to at most $\lambda - 1$, the product of the exact factors of a summand is its exact value times $1 + \theta$ with $|\theta| \leq \gamma_{\nu(\lambda-1)}$. The summand then undergoes at most ν roundings: at most one in the product, at most one in each addition to the running sum of its group, and at most one in each addition of a group sum. Therefore the computed value is $\sum_i z_i(1 + \theta_i)$, where the $z_i \geq 0$ are the exact summands and $|\theta_i| \leq \gamma_{\nu\lambda}$. This is $(1 + \theta) \sum_i z_i$ with $|\theta| \leq \gamma_{\nu\lambda}$. $\square$

Proposition A.2 (Floating-point sampling). *Let $n \leq 300$ and $g = \gamma_{\nu N}$. Assume that the random bits used by the sampler are independent and uniform. Then the output of the floating-point sampler is within total variation distance*

$$n \left(\frac{g}{1 - g} + 7\nu u \right) < 3.5 \cdot 10^{-5}$$

of the uniform distribution on $\text{Av}_n(12453)$.

Proof. By [lemma A.1](#) the sampler reaches only states of positive count. Fix such a state, with stored value $\hat{T}$ and candidates $1, \dots, k$ in scan order. Let w_i be the scaled exact value of candidate i and $W = w_1 + \dots + w_k$. Let v_i be the exact product of the stored values in candidate i and its power of two, and let $\hat{w}_i = \text{fl}(v_i)$ be the value the sampler computes, with $\hat{w}_i = v_i$ when there is one factor.

The values. By [lemma A.1](#) applied to the factors, $\hat{w}_i = w_i(1 + \theta_i)$ with $|\theta_i| \leq \gamma_{\nu(\lambda-1)+1} \leq g$, and $\hat{T} = W(1 + \theta)$ with $|\theta| \leq g$. Therefore

$$\sum_{i=1}^k \left| \frac{\hat{w}_i}{\hat{T}} - \frac{w_i}{W} \right| = \sum_{i=1}^k \frac{w_i}{W} \left| \frac{1 + \theta_i}{1 + \theta} - 1 \right| \leq \frac{2g}{1 - g}. \quad (44)$$

The stored value $\hat{T}$ and the total $\hat{A}$ of the scan are both computed sums of $v_1, \dots, v_k$ in which each v_i undergoes at most ν roundings. The two endpoint candidates are one summand of the table, of value $v_i + v_{i+1}$. Therefore $\hat{T}$ and $\hat{A}$ are both within a factor $1 \pm \gamma_\nu$ of $v_1 + \dots + v_k$, and $|\hat{A} - \hat{T}| \leq 2.01\nu u \hat{T}$.

The scan. For $y < \hat{A}$ the chosen candidate is a nondecreasing function of y , since rounding is monotone. So candidate i is chosen for y in an interval $[\beta_{i-1}, \beta_i)$, where $0 = \beta_0 \leq \dots \leq \beta_k = \hat{A}$. Consider a group with running sums $a_1 \leq \dots \leq a_r$ of its members, and let A_{g-1} and A_g be the running sums before and after it. The group occupies $[A_{g-1}, A_g)$, and its member $j < r$ ends where $\text{fl}(y - A_{g-1})$ reaches a_j , or at A_g if that comes first. Since $|\text{fl}(z) - z| \leq u|z|$, this breakpoint differs from $A_{g-1} + a_j$ by at most $1.01u\hat{A}$. The difference $a_j - a_{j-1}$ differs from the value of member j by at most $u\hat{A}$, and $A_g - A_{g-1}$ differs from a_r by at most $u\hat{A}$. Hence

$$|\beta_i - \beta_{i-1} - \hat{w}_i| \leq 4.02u\hat{A} \quad (1 \leq i \leq k). \quad (45)$$

The draw. For $\tau \geq 0$ let $F(\tau)$ be the probability that $y \geq \tau$, and let $f(\tau) = \max(0, 1 - \tau/\hat{T})$. Since $x\hat{T} < \hat{T}$, we have $y \leq \hat{T}$, so $F(\tau) = 0 = f(\tau)$ for $\tau > \hat{T}$. For $\tau \leq \hat{T}$, we have $y \geq \tau$ if $x\hat{T} \geq \tau/(1-u)$ and only if $x\hat{T} \geq \tau/(1+u)$. Since x is uniform on the multiples of 2^{-53} in $[0, 1)$, the probability that $x \geq z$ lies between $1 - z - u$ and $1 - z$. Together,

$$|F(\tau) - f(\tau)| \leq 2.01u \quad (0 \leq \tau \leq \hat{A}). \quad (46)$$

One letter choice. Let $\hat{p}_i$ be the probability that the sampler chooses candidate i . Without a rescan, candidate i is chosen with probability $F(\beta_{i-1}) - F(\beta_i)$. By (46), replacing F by f changes the sum over i of these probabilities by at most $4.02ku$ in total. The differences $f(\beta_{i-1}) - f(\beta_i)$ equal $(\beta_i - \beta_{i-1})/\hat{T}$ except above $\hat{T}$, which changes the sum by at most $(\hat{A} - \hat{T})/\hat{T} \leq 2.01\nu u$. By (45), replacing $\beta_i - \beta_{i-1}$ by $\hat{w}_i$ changes it by at most $4.03ku$. A rescan happens with probability $F(\hat{A}) \leq f(\hat{A}) + 2.01u \leq 2.01\nu u + 2.01u$. Since $k < \nu$,

$$\sum_{i=1}^k \left| \hat{p}_i - \frac{\hat{w}_i}{\hat{T}} \right| \leq 8.05ku + 4.02\nu u + 2.01u \leq 14\nu u.$$

With (44), the total variation distance between the choice of the sampler and the choice with probabilities w_i/W is at most $g/(1-g) + 7\nu u$.

The sample. With probabilities w_i/W , the choices produce the uniform distribution on $\text{Av}_n(12453)$, by [proposition 8.1](#). Run the two processes together, and as long as they are in the same state, couple their next choices so that they differ with probability equal to the total variation distance between them. A sample is produced by n choices, so the two outputs differ with probability at most $n(g/(1-g) + 7\nu u)$. At $n = 300$ this is $3.43 \cdot 10^{-5}$. $\square$

The bound of [lemma A.1](#) is far from the observed error: at $n = 150$ it gives about $6 \cdot 10^{-8}$ for $G_{(n,0)}$, against the observed 10^{-11} .

APPENDIX B. THE LEAN DEVELOPMENT

This appendix describes the Lean development of [section 9](#).

B.1. Conventions. The development works with words over $\mathbb{N}$ and 0-based values. It defines pattern containment as the existence of an order-isomorphic subsequence, a finite set of 0-based words in bijection with $\text{Av}_n(\tau)$, the one-threshold recurrence W of (5), the literal recurrence H of (22) for $d = 2$, and the kernel recurrences (13)–(15) for $d = 1$ and (27)–(29) for $d = 2$. In the two kernel systems every sum over controls is restricted to the controls of mass at most $\|\mathbf{p}\|_1$ (to $\{0, \dots, p\}$ for $d = 1$), which [lemmas 2.20](#) and [6.1](#) show loses nothing. The four main

theorems of [section 9](#) are named `av1342_count`, `av1342_count_kernel`, `av12453_count`, and `av12453_count_kernel`.

B.2. Organization. The Lean project `formal/Av12453/` builds two libraries from one source tree. The library `PermPatterns/` holds the generic apparatus in ten modules: words and order isomorphism, containment and avoidance, restriction, local rank and standardization, the patterns 231 and ι_d with increasing subsequences and triggers, direct and skew sums, decidability of containment, the finite sets of permutations and of avoiders of a basis, the passage between words and Mathlib’s permutations of $\{0, \dots, n-1\}$, the symmetries of reversal, complementation, and inversion with the equalities of avoider counts they induce, and [lemma 2.2](#). Nothing in it refers to the family β_d . The library `Av12453/` holds the development specific to this paper in twenty modules. The modules `Basic.lean` and `Trigger.lean` define the patterns β_d and prove [lemma 2.1](#). The eight modules of `OneThreshold/` treat $d = 1$. The module `Defs.lean` defines the scan, its state (p, L) , and W . The modules `Invariant.lean`, `Semantics.lean`, and `Counting.lean` prove the separation invariant of [lemma 2.11](#), parts (b) and (c) of [proposition 2.10](#), and `av1342_count`. The modules `Kernel.lean`, `KernelSupport.lean`, `KernelFactor.lean`, and `KernelCount.lean` define K , D , and G , and prove [lemma 2.20](#), the factorization (12), and `av1342_count_kernel`. The nine modules of `TwoThreshold/` repeat this for $d = 2$, with `Thresholds.lean` for the two thresholds and their bands ([section 3](#)) and `Defs.lean` for H . The directory also contains `Axioms.lean`, which walks every constant declared in every module of the two libraries and fails if any depends on `sorryAx` or on an axiom other than propositional extensionality, choice, and quotient soundness. The sweep finds its list of modules in the environment rather than in a hand-written list, so a module cannot drop out of it silently. The script `check.sh` builds the project, requires the build to be free of warnings, runs the axiom sweep, and rejects any occurrence of `sorry` or `native_decide` in the sources. There is also a `README.md` with a table of the modules and their headline theorems, and an audit report. The Lean toolchain (version 4.33.1) and the Mathlib revision are pinned in the project files.

B.3. What must be read. The first of the two tasks of [section 9](#) is to run `check.sh`, or equivalently `lake build` followed by the axiom sweep, and see it end with the line `CERTIFICATE CHECK PASSED`. The second is to read, against this paper, the definitions that occur in the four theorem statements. Nothing else needs to be read to know what the theorems say. They are listed by file. In the library `PermPatterns/`, the module `Word.lean` defines `IsPermOf` and `OrderIsomorphic`, `Containment.lean` defines `Contains`, `Avoids`, and `AvoidsAll`, `Patterns.lean` defines `iota`, and `Avoiders.lean` defines `perms` and `avoiders`. In the library `Av12453/`, the module `Basic.lean` defines `beta`, so that `beta 2` is the 0-based word 0, 1, 3, 4, 2 of 12453, the module `OneThreshold/Defs.lean` defines `nz` and `W`, and `TwoThreshold/Defs.lean` defines `H`. The module `OneThreshold/Kernel.lean` defines K , D , and G for $d = 1$, and `TwoThreshold/Kernel.lean` defines the control sets `ctrls` and `S` and K , D , and G for $d = 2$. These definitions occupy about seventy lines of Lean in all. The recurrences are written with a recursion bound. The definitions `W`, `H`, `K`, and `G` evaluate an auxiliary function at a bound exceeding the grade of the argument, and the development proves that every sufficient bound gives the same value. So the reader need only compare the equations of the auxiliary functions with the displays (5), (22), (13)–(15), and (27)–(29). Each definition carries a docstring giving this dictionary, with the 0-based conventions spelled out, and the displays are also restated as proved theorems (`K_succ_eq`, `G_eq`, and their analogues), which serve as reading aids but are not part of what must be trusted.

B.4. Proofs and checks. In more detail, the state of a scanned prefix is defined from the prefix: the thresholds b_1, b_2 as the least final values of increasing subsequences, the bands, and the interval stack by the legal moves of [definition 2.4](#), justified by [lemmas 2.2](#), [2.5](#) and [2.8](#).

The inductive invariant is the separation criterion of [lemma 2.11](#) and [corollary 4.4](#), from which two facts are derived, [proposition 2.10\(b\)](#) and (c): a letter from a deferred interval leaves no avoiding completion, and every complete word produced by legal moves avoids β_d . The completion counts then satisfy the literal recurrence by induction on the number of unread values, and the factorization (25) is proved directly from the equations of H by induction on the grade, without the abstract [theorem 2.17](#). The kernels are defined by their recurrences, so the theorems concern the recurrences as displayed. The restriction of the control sums is proved lossless, and it is also what makes the grade a well-founded order for (27) read as a definition.

The development has over eleven thousand lines in thirty modules, contains no unproved statement, and each theorem depends only on Lean’s three standard axioms, as reported by the axiom sweep. The definitions were also exercised by kernel evaluation: the scans of the running example (4) reproduce the rows of [examples 2.12](#) and [4.2](#), kernel evaluation of the recurrences reproduces the first seven terms of (2), and compiled evaluation the first ten.

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